A Project Report on

SMART HEALTHCARE SYSTEM USING AI & BLOCKCHAIN

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

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We hereby declare that this project entitled " SMART HEALTHCARE SYSTEM USING AI & BLOCKCHAIN " has been undertaken by us, and this work has been submitted to ADITYA COLLEGE OF ENGINEERING & TECHNOLOGY (A), Surampalem, affiliated to JNTUK, Kakinada, in partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING.

We further declare that this project work has not been submitted in full or in part to any other University or educational institute for the award of any degree or diploma.

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ABSTRACT

The Smart Healthcare System using AI and Blockchain for Heart Disease is designed to support early diagnosis, secure medical record management, and remote healthcare services for diverse patient groups. The system analyzes patient health parameters using a machine learning model to predict the risk of heart disease, enabling timely preventive care and clinical decision support. It dynamically links prediction results with a blockchain-based ledger, ensuring that medical reports are tamper-evident and verifiable by authorized users such as doctors, patients, and administrators. Through an integrated telemedicine interface, patients can securely access their reports, consult doctors, and receive follow-up guidance without needing frequent hospital visits.

By combining AI-driven prediction, blockchain-secured records, and telemedicine, the platform enhances accuracy, transparency, and accessibility in healthcare delivery. The system continuously records each prediction as a hashed ledger entry, preserving an auditable history of medical reports and increasing trust in digital health services.

A modern web interface built with React and a Python–Flask backend allows seamless interaction between users, the Decision Tree classifier, the ledger table, and the PostgreSQL database. This integrated approach creates a practical smart healthcare ecosystem that addresses challenges such as delayed diagnosis, insecure data handling, and limited access to quality care in remote areas.

KEYWORDS: Smart Healthcare, Heart Disease Prediction, Artificial Intelligence, Machine Learning, Blockchain, Telemedicine, Secure Medical Records, Decision Tree Classifier.

TABLE OF CONTENTS

CONTENT	PAGE NO
ABSTRACT	VII
LIST OF FIGURES	XII
LIST OF TABLES	XIV
1 INTRODUCTION	1-7
1.1 Introduction to Project	2
1.2 Literature Survey	3
1.3 Existing System	3
1.4 Scope of the Existing System	3
1.5 Proposed System	4-5
1.6 Novelty of the Proposed System	6-7
2 LITERATURE SURVEY	8-15
2.1 Introduction to Literature Survey	9
2.2 Theoretical Background/Key Concepts	9
2.3 Existing Methods / Systems	9-11
2.3.1 AI-Based Disease Prediction Systems	9-10
2.3.2 Blockchain-Based Healthcare Systems	10
2.3.3 Telemedicine Consultation Systems	11
2.4 Scope of Existing Systems	12-13
2.4.1 Comparative Analysis	12
2.4.2 Critical Limitations Identified	12-13
2.4.3 Research Group	13
2.5 Proposed System	14-15
2.5.1 Key Findings Summary	14
2.5.2 How the Proposed System Identifies Gaps	14-15
2.5.3 Theoretical Contribution	15
2.5.4 Practical Implications	15

3 REQUIREMENT ANALYSIS 16-21
3.1 Functional Requirements 17-18
3.2 Non-Functional Requirements 19-21
3.2.1 User Interface Requirements 19
3.2.2 Software Requirements 19
3.2.3 Hardware Requirements 19-20
3.2.4 Performance Requirements 20
3.2.5 Usability Requirements 20
3.2.6 Reliability Requirements 20
3.2.7 Scalability Requirements 20-21
4 SYSTEM DESIGN 22-40
4.1 Overview of System Architecture 23
4.2 Module Description / Decomposition 23-24
4.3 Database Design 24
4.4 Data Flow / Control Flow 25
4.5 UML Diagrams 26-39
4.5.1Use Case Diagram 26
4.5.2Class Diagram 27
4.5.3Sequence Diagram 28
4.5.4Activity Diagram 29
4.5.5State Diagram 30
4.5.6Component Diagram 31
4.5.7 Deployment Diagram 32
4.5.8 Package Diagram 33
4.5.9 Composite Structure Diagram 34
4.5.10 Profile Diagram 35
4.5.11 Communication Diagram 36
4.5.12 Interaction Overview Diagram 37
4.5.13 Timing Diagram 38
4.5.14 Additional UML Diagram 39

4.6 Design Decisions and Constraints 40
5 SYSTEM IMPLEMENTATION 41-55
5.1 Implementation Environment 42-43
5.2 Module-wise Implementation 43-47
5.2.1 User Authentication Module 43-44
5.2.2 Data Processing Module 44-45
5.2.3 Prediction / Analysis Module 45
5.2.4 Blockchain/Ledger Module 46
5.2.5 Telemedicine Module 47
5.3 Algorithm / Pseudo Code 48-49
5.4 Screenshots and Explanation 50-55
6 TESTING AND VALIDATION 56-64
6.1 Testing Strategy 57-60
6.1.1 Unit Testing 57-58
6.1.2 Integration Testing 58-59
6.1.3 System Testing 59-60
6.2 Test Cases 60
6.3 Test Results 62
6.4 Validation 52-64
7 RESULTS AND PERFORMANCE ANALYSIS 65-67
7.1 Output Analysis 66
7.2 Performance Metrics 66
7.3 Comparative Analysis 67
8 CONCLUSION AND FUTURE ENHANCEMENTS 68-71
8.1 Conclusion 69-70
8.2 Limitations 70
8.3 Future Scope / Enhancements 71
9 REFERENCES 72-74
10 APPENDIX 75-90
10.1 Source Code (Sample / Snippets) 76-77

10.2 User Manual 77-78
10.3 Abbreviations 78
10.4 Publication Paper Copy 79-85
10.5 Publication Certificate 86-90

LIST OF FIGURES

SNO	FIGURE NO	FIGURE NAME	PAGE NO
1.	4.1.1	System Architecture	23
2.	4.3.1	Database Design	24
3.	4.4.1	Data-Flow Diagram	25
4.	4.5.1.1	Use Case Diagram	26
5.	4.5.2.1	Class Diagram	27
6.	4.5.3.1	Sequence Diagram	28
7.	4.5.4.1	Activity Diagram	29
8.	4.5.5.1	State Diagram	30
9.	4.5.6.1	Component Diagram	31
10.	4.5.7.1	Deployment Diagram	32
11.	4.5.8.1	Package Diagram	33
12.	4.5.9.1	Composite Structure Diagram	34
13.	4.5.10.1	Profile Diagram	35
14.	4.5.11.1	Communication Diagram	36
15.	4.5.12.1	Interaction Overview Diagram	37
16.	4.5.13.1	Timing Diagram	38
17.	4.5.14.1	Additional UML Diagram	39
18.	5.4.1	Patient/Doctor Login	50
19.	5.4.2	AI Disease Prediction	51
20.	5.4.3	Blockchain record verification	52
21.	5.4.4	Appointment Booking	53
22.	6.2.2	Prediction & Hashing Tests	61

23.	10.1.1	Source Code	77
24.	10.4	Publication Paper copy	79-85
25.	10.5	Publication Certificate	86-90

LIST OF TABLES

S NO.	TABLE NO.	TABLE NAME	PAGE NO.
1.	2.4.1.1	Comparative Analysis	12
2.	6.2.1	Test Case for the Functionality of the System	60
3.	7.3.1	Comparing Features between Existing and Proposed Systems	67
4.	10.3.1	List of Abbreviations used in the Project	72

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

1.1 Introduction to Project:

The healthcare system, especially in developing regions, continues to face challenges such as delayed diagnosis, fragmented medical records, and limited access to specialist doctors. Patients often seek medical attention only after symptoms become severe, while doctors may lack complete and reliable historical data for accurate decision-making.

To address these issues, this project proposes a **Smart Healthcare System using Artificial Intelligence (AI) and Blockchain technology for Heart Disease Prediction**. The system enables patients to input health parameters such as age, blood pressure, cholesterol levels, and other medical attributes, which are analyzed using machine learning algorithms to predict the risk of heart disease at an early stage.

In addition, the system ensures secure and tamper-proof storage of medical reports using blockchain technology, enhancing the trustworthiness and integrity of healthcare data. It also includes a telemedicine interface that facilitates remote interaction between patients and doctors, improving accessibility to quality healthcare services.

Overall, this integrated approach aims to improve diagnosis accuracy, enable early disease detection, ensure secure medical data management, and provide efficient, reliable, and accessible digital healthcare solutions.

1.2 Literature Survey

Several research studies highlight the importance of Artificial Intelligence and machine learning technologies in improving disease prediction and healthcare services. Studies indicate that AI-

based diagnostic systems can analyze patient medical data and help in the early detection of diseases such as heart disease, enabling doctors to take preventive measures in time. Blockchain technology also plays an important role in ensuring secure storage and integrity of medical records. Existing research confirms that AI-driven healthcare systems improve diagnosis accuracy, support medical professionals in decision making, and enhance patient care. However, many current systems focus only on disease prediction and lack secure and transparent data management, which the proposed system addresses by integrating AI-based prediction with blockchain-based medical record security.

1.3 Existing System

In the existing healthcare system, many hospitals and clinics rely on centralized databases or manual record keeping for storing patient medical information. These systems are often vulnerable to data tampering, unauthorized access, and loss of important medical records. Patients usually need to visit hospitals physically to receive medical consultation and diagnosis, which may cause delays in identifying serious health conditions such as heart disease.

Traditional healthcare systems depend on paper-based records, manual diagnosis, and limited digital support. Existing disease prediction systems mainly focus on analyzing medical data but do not provide secure storage or verification of medical reports. Many healthcare platforms also lack integrated telemedicine services, making it difficult for patients to access healthcare services remotely.

1.4 Scope of the Existing System

The scope of the existing healthcare system is limited due to the absence of secure medical record management and intelligent disease prediction technologies. Most healthcare systems focus only on diagnosis and treatment after patients visit hospitals, rather than supporting early detection and preventive healthcare.

Existing systems mainly rely on centralized databases, which increases the risk of data manipulation and security breaches. Medical records stored in traditional systems are difficult to verify and track over time, which can affect the reliability of patient information.

There is also limited integration of advanced technologies such as Artificial Intelligence and Blockchain for predictive healthcare and secure data management. As a result, the existing systems fail to provide a secure, transparent, and efficient healthcare environment, highlighting the need for a technology-driven solution like the Smart Healthcare System.

1.5 Proposed System

The proposed system, **Smart Healthcare System using AI and Blockchain for Heart Disease Prediction**, is designed to improve healthcare services by integrating Artificial Intelligence and Blockchain technologies. The system analyzes patient health parameters using machine learning algorithms to predict the risk of heart disease at an early stage.

Once the prediction is generated, the medical report is secured using blockchain-based hashing, ensuring that the stored data is tamper-proof and trustworthy. The system also provides a telemedicine interface that allows patients to access their reports and consult doctors remotely. This approach aims to improve early disease detection, enhance data security, and provide reliable digital healthcare services.

Key Features of the Proposed System:

1. **Web-Based User Interface:**

   Users can register, log in, and enter their health parameters such as age, blood pressure, cholesterol level, and other medical details through a simple web-based interface.

2. **AI-Based Heart Disease Prediction:**

   The system uses a machine learning model to analyze patient health data and predict the risk of heart disease, helping in early detection and preventive care.

3. **Blockchain-Based Record Security:**

   Medical reports are secured using blockchain-style hashing, ensuring that the stored data is tamper-proof and cannot be modified without detection.

4. **Doctor Access and Report Verification:**

   Doctors can access patient prediction reports and verify their authenticity using transaction IDs or QR codes linked to the blockchain ledger.

5. **Prediction Result Dashboard:**

   Provides a clear interface where patients can view their prediction results, health risk levels, and previously generated reports.

6. **Secure Medical Data Management:**

   Stores patient health information and prediction history in a structured database for easy access and monitoring.

7. **Telemedicine Support:**

   Allows patients to consult doctors remotely and share their verified medical reports for better diagnosis and treatment.

8. **QR Code Verification System:**

   Each prediction report is linked to a unique QR code that allows easy verification of the report's authenticity and integrity.

1.6 Novelty of the Proposed System

The **Smart Healthcare System using AI and Blockchain for Heart Disease Prediction** introduces several innovative features that distinguish it from existing healthcare solutions. Its novelty lies in combining Artificial Intelligence and Blockchain technology to provide accurate disease prediction along with secure and tamper-proof management of medical records. This integrated approach improves early diagnosis, enhances data security, and supports reliable digital healthcare services.

Key Innovative Aspects:

1. **AI-Based Heart Disease Prediction:**

   Unlike traditional systems, the proposed system analyzes patient health parameters using machine learning algorithms to predict the risk of heart disease and support early medical diagnosis.

2. **Blockchain-Based Medical Record Security:**

   The system stores prediction reports using blockchain-style hashing, ensuring that medical data remains tamper-proof and secure from unauthorized modifications.

3. **QR Code and Transaction ID Verification:**

   Each medical report is linked with a unique QR code or transaction ID that allows doctors and authorized users to verify the authenticity of the report.

4. **Secure Digital Health Records:**

   Patient medical records and prediction history are securely stored and can be accessed by authorized users, ensuring continuity of care and easy retrieval of past medical data.

5. **Telemedicine Integration:**

   The system enables patients to share their prediction reports with doctors and receive medical consultation remotely, improving accessibility and convenience.

Impact and Benefits:

The innovative combination of AI diagnostics, telemedicine, and cloud-based management empowers rural populations with accessible and affordable healthcare, improves early detection of diseases, enhances the efficiency of health workers, and significantly reduces the healthcare disparity between rural and urban areas.

CHAPTER 2

LITERATURE SURVEY

2. Literature Survey

2.1 Introduction to literature survey

The literature survey chapter reviews existing research and systems related to AI-based heart disease prediction and blockchain-enabled medical record management. It helps to understand current techniques, datasets, and technologies used in healthcare applications and clarifies how our work fits into that context.

This review also highlights limitations of existing systems in terms of data security, transparency, and accessibility. By identifying clear research gaps, the survey provides a strong motivation for developing an integrated smart healthcare system using AI, blockchain, and telemedicine.

2.2 Theoretical Background / Key Concepts

- **Artificial Intelligence & Machine Learning:** Analyze patient health data to predict heart disease risk automatically.
- **Decision Tree Classifier:** A supervised model that uses rule-based splits to classify patients into disease or no-disease.
  Blockchain: Stores medical records using cryptographic hashes to ensure tamper-proof and secure data integrity.
- **Private Permissioned Blockchain:** Allows access only to authorized users like doctors and patients for confidentiality.
- **Telemedicine:** Enables online consultation, report access, and appointment booking for remote healthcare services.

2.3 Review of Existing Methods and Systems

2.3.1 AI-Based Disease Prediction Systems

1. **Heart Disease Prediction using Machine Learning**

   • Focus: Predicting heart disease risk using patient medical parameters.

• Methods: Algorithms such as Decision Tree, Logistic Regression, and Random Forest are applied on healthcare datasets.

• Results: These systems help doctors in early detection of cardiovascular diseases.

• Limitations: Most systems focus only on prediction and do not provide secure storage of medical records.

2. **Clinical Decision Support Systems (CDSS)**

• Focus: Assisting doctors in medical diagnosis and treatment planning.

• Methods: Machine learning models analyze patient data and generate diagnostic suggestions.

• Results: Improves decision-making and reduces human errors in diagnosis.

• Limitations: Requires centralized databases and lacks strong data security mechanisms.

2.3.2 Blockchain-Based Healthcare Systems

3. **Blockchain for Medical Record Management**

• Focus: Secure storage and sharing of patient medical records.

• Methods: Blockchain networks store patient data using cryptographic hashing and distributed ledgers.

• Results: Ensures tamper-proof storage and improves transparency in healthcare systems.

• Limitations: Many systems focus only on data security and do not integrate AI-based disease prediction.

4. **Blockchain-Based Electronic Health Records (EHR)**

• Focus: Providing decentralized and secure access to patient health records.

• Methods: Smart contracts and cryptographic verification techniques are used for secure data sharing.

• Results: Improves trust and data integrity in digital healthcare systems.

• Limitations: Implementation complexity and lack of integration with predictive healthcare models.

2.3.3 Telemedicine and Digital Healthcare Platforms

5. **Telemedicine Consultation Systems**

   • Focus: Providing remote medical consultation services to patients.

   • Methods: Web and mobile applications that support video calls, messaging, and appointment booking.

   • Results: Improves accessibility to healthcare services in remote areas.

   • Limitations: Most systems do not ensure secure verification of medical reports.

6. **Online Healthcare Monitoring Systems**

   • Focus: Continuous monitoring of patient health data using digital platforms.

   • Methods: Sensors, cloud storage, and healthcare management systems are used for monitoring patient data.

   • Results: Enables doctors to track patient health conditions remotely.

   • Limitations: Data security and authenticity of medical reports are often not guaranteed.

2.4 Critical Analysis and Research Gaps

2.4.1 Comparative Analysis

System	Strengths	Limitations	Suitability
AI-based Prediction Models	Accurate disease prediction using medical datasets	Lack secure medical record storage	Moderate
Blockchain Healthcare Systems	Secure and tamper-proof data storage	Limited support for disease prediction	Moderate
Telemedicine Platforms	Remote doctor consultation and accessibility	Weak data security and verification mechanisms	Good
Traditional Hospital Systems	Reliable medical diagnosis by experts	Manual record delayed diagnosis	Limited

Table no. 2.4.1.1 Comparative analysis

2.4.2 Critical Limitations Identified

1. Data Security Issues

• Many healthcare systems rely on centralized databases that are vulnerable to unauthorized access.

• Medical records can be modified without proper verification.

• Lack of transparency in maintaining patient medical data.

2. Limited Disease Prediction Support

• Most healthcare platforms depend on manual diagnosis by doctors.

• Early prediction of diseases using AI is not widely implemented.

3. Lack of Integrated Healthcare Systems

• Existing systems focus only on one feature such as prediction or consultation.

• There is limited integration between AI-based analysis, secure data storage, and healthcare services.

4. Accessibility Challenges

• Many healthcare services require physical hospital visits.

• Remote areas may not have easy access to medical specialists.

2.4.3 Research Gaps

Gap 1: Integration of AI and Blockchain

• Existing systems focus either on disease prediction or secure record management, but rarely integrate both technologies.

Gap 2: Secure Verification of Medical Reports

• Many digital healthcare systems do not provide mechanisms to verify the authenticity of medical reports.

Gap 3: Early Detection of Heart Disease

• Traditional healthcare systems detect heart disease only after symptoms become severe.

Gap 4: Secure and Transparent Healthcare Data Management

• There is a need for systems that ensure data integrity, transparency, and trust in medical records.

2.5 Summary and Link to Proposed Work

2.5.1 Key Findings Summary

1. Importance of Early Disease Prediction:

   • Machine learning techniques are effective in predicting heart disease using patient medical data.
   • Early prediction helps doctors take preventive measures and improve patient outcomes.

2. Need for Secure Medical Data Management:

   • Healthcare systems require strong security mechanisms to protect sensitive patient records.
   • Blockchain technology provides secure and tamper-proof storage of medical data.

3. Role of Digital Healthcare Platforms:

   • Online healthcare platforms and telemedicine improve accessibility to medical services.

   • Digital systems allow patients and doctors to share health information more efficiently.

4. Integration of Advanced Technologies:

   • Combining Artificial Intelligence with secure data management technologies can improve healthcare systems.
   • Integrated solutions provide better diagnosis support and reliable healthcare services.

2.5.2 How the Proposed System Addresses Identified Gaps

Addressing Gap 1: AI-Based Disease Prediction

• Approach: Use machine learning algorithms to analyze patient health parameters.

• Innovation: Early prediction of heart disease risk using medical datasets.

• Implementation: Decision Tree model trained on heart disease datasets.

Addressing Gap 2: Secure Medical Record Management

• Approach: Use blockchain-based hashing to secure medical reports.

• Innovation: Tamper-proof storage and verification of patient reports.

• Implementation: SHA-256 hashing with blockchain-style ledger storage.

Addressing Gap 3: Digital Healthcare Accessibility

• Approach: Provide a web-based healthcare platform for patient interaction.

• Innovation: Patients can access their prediction reports and share them securely with doctors.

• Implementation: Web application developed using modern frontend and backend technologies.

2.5.3 Theoretical Contribution

The proposed system contributes to healthcare technology research by:

1. Integrating AI-based heart disease prediction with blockchain-secured medical records.
2. Improving data transparency and integrity in digital healthcare systems.
3. Demonstrating how machine learning can support early disease detection.
4. Providing a secure framework for managing healthcare data.
5. Combining predictive analytics with digital healthcare platforms.

2.5.4 Practical Implications

The literature survey confirms that the proposed system can improve healthcare services by:

• Supporting early detection of heart disease through AI-based prediction.

• Protecting patient medical records using blockchain security.

• Improving data reliability and transparency in healthcare systems.

• Allowing secure sharing of medical reports between patients and doctors.

• Providing a technology-driven healthcare platform for better medical decision making.

CHAPTER 3

REQUIREMENT ANALYSIS

3. Requirement Analysis

3.1 Functional Requirements

FR-001: User Registration and Login

• The system shall allow users (patients and doctors) to register and log in using secure authentication.
• The system shall allow patients to enter their personal and medical details.

FR-002: Patient Health Data Entry

• The system shall allow patients to enter health parameters such as age, blood pressure, cholesterol level, chest pain type, and other medical attributes.
• The system shall validate the entered data before processing.

FR-003: AI-Based Heart Disease Prediction

• The system shall analyze patient health parameters using a Decision Tree machine learning model.
• The system shall generate a prediction indicating whether heart disease risk is present or not along with a risk category.

FR-004: Blockchain-Based Medical Record Storage

• The system shall generate a SHA-256 cryptographic hash of each prediction report.

• The hash shall be stored in a blockchain-style ledger table to ensure data integrity and tamper-proof records.

FR-005: Report Verification System

• The system shall generate a QR code or transaction ID for each prediction report.

• The system shall allow users to verify the authenticity of reports using the verification page.

FR-006: Telemedicine Support

• The system shall allow doctors to log in and view patient prediction reports.

• Doctors shall be able to provide consultation notes or recommendations based on the prediction results.

FR-007: Patient Record Management

• The system shall store patient health records and prediction history in a database.

• Patients shall be able to view their previous reports and health data.

FR-008: Administrative Control

• The system shall allow administrators to manage users and monitor prediction transactions. Administrators shall be able to oversee system configuration and platform activity.

3.2 Non-Functional Requirements

3.2.1 User Interface Requirements

The system interface should be simple, responsive, and easy to use so that both technical and non-technical users can interact with the platform without difficulty.

• The interface shall provide separate dashboards for patients, doctors, and administrators.

• Forms and dashboards shall display prediction results clearly.

• The interface shall support responsive design for desktop and mobile devices.

3.2.2 Software Requirements

The software technologies required for the system include:

• **AI/ML Tools:** Python, Scikit-learn, Pandas, NumPy for developing the heart disease prediction model.

• **Backend Framework:** Flask for building APIs and integrating AI, blockchain logic, and database services.

• **Frontend Technologies:** React.js with TypeScript and Tailwind CSS for building the web interface.

• **Database: PostgreSQL** for storing patient records, predictions, and ledger transactions.

• **Blockchain Logic:** SHA-256 hashing with ledger table to provide tamper-proof verification.

• **Development Tools:** VS Code, Postman, and GitHub for coding, testing, and version control.

3.2.3 Hardware Requirements

The system can operate on standard computing devices.

• A computer or laptop capable of running a web browser and backend services.

• Minimum 8 GB RAM recommended for smooth system performance.

• Stable internet connection required for telemedicine and online verification.

• At least 2–3 GB storage for application files and databases.

3.2.4 Performance Requirements

• The heart disease prediction module should generate results within a few seconds after the user submits health data.

• Database operations and blockchain ledger updates should handle multiple transactions efficiently.

• The system should support concurrent users without significant delay.

3.2.5 Usability

The system should be easy to learn and operate.

• Patients should be able to submit health data and view predictions in a few steps.

• Doctors should be able to view reports and provide consultation easily.

• Error messages and validation should guide users when incorrect data is entered.

3.2.6 Reliability

The system should maintain high data reliability and integrity.

• Blockchain-style hashing ensures medical reports cannot be modified without detection.

• The system should handle failures gracefully with proper error handling.

• Regular database backups should be maintained to prevent data loss.

3.2.7 Scalability

• The platform should support increasing numbers of patients and doctors.

• The system architecture should support future expansion.

- Application and database components can be deployed on separate servers if required.
- The ledger system can be extended for cross-hospital medical data sharing in the future.

CHAPTER-4

SYSTEM DESIGN

4. System Design

4.1 Overview of System Architecture

The system follows a centralized, web-based layered architecture. Patients and doctors interact with the system through a browser-based React application. The Flask backend manages authentication, API communication, prediction processing, and database operations. The AI module analyzes patient health parameters using a Decision Tree model to predict heart disease risk. After prediction, a SHA-256 hash of the medical report is generated and stored in a blockchain-style ledger to ensure tamper-proof and secure records. The PostgreSQL database stores patient details, prediction history, and transaction logs, enabling secure access and verification through QR codes or transaction IDs.

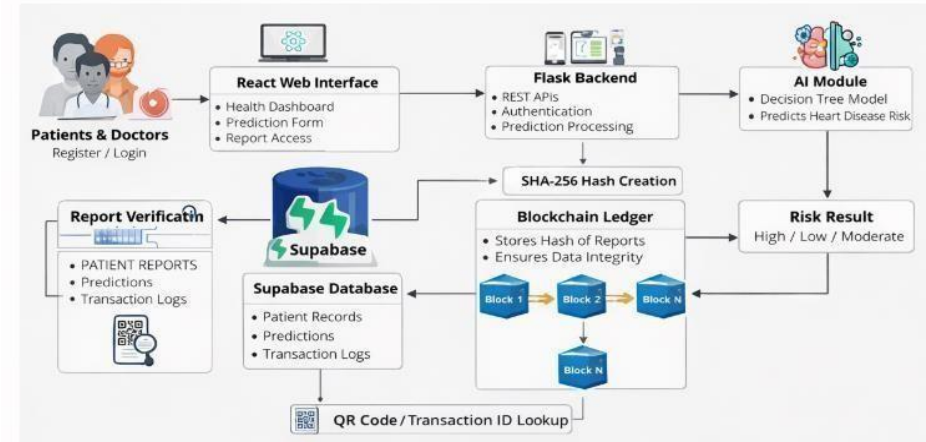


Fig no. 4.1.1System Architecture

4.2 Module description/decomposition

- **Patient Module:** Handles patient registration, login, profile management, and input of health parameters for prediction. Output includes prediction results and access to past reports.
- **Doctor Module**: Allows doctors to log in, view patient reports, verify report integrity via blockchain, and record consultation notes or recommendations.
- **AI Prediction Module**: Receives preprocessed medical attributes as input, applies the Decision Tree classifier, and returns a disease/no disease prediction with risk level.
- **Blockchain / Ledger Module**: Converts key report details into a hash, stores it in a ledger

table with links to previous hashes, and supports QR-based verification.

- **Telemedicine Module**: Supports basic communication and appointment or consultation workflows between patients and doctors, integrated with report viewing.**Admin Module:** Enables administrators to manage user accounts, monitor transactions, and configure system parameters.

4.3 Database design

The database consists of tables for users, patient records, predictions, ledger entries, and appointments. Users manage authentication and roles, while patient and prediction tables store health data and results with defined relationships. The ledger maintains hashes and timestamps linked to predictions, ensuring data integrity through a secure chain.

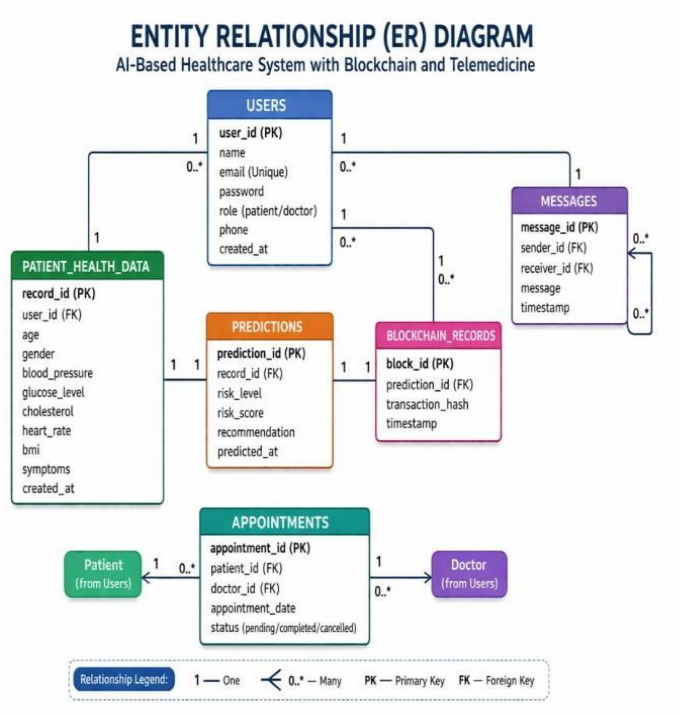


Fig no. 4.3.1Databse design

4.4 Data flow/control flow

Data flow begins when a patient submits health data, which is validated, processed, and analyzed by the AI model to generate a prediction stored in the database. A hash and QR/ID are created for the report, enabling doctors to verify data integrity by matching hashes before accessing results.

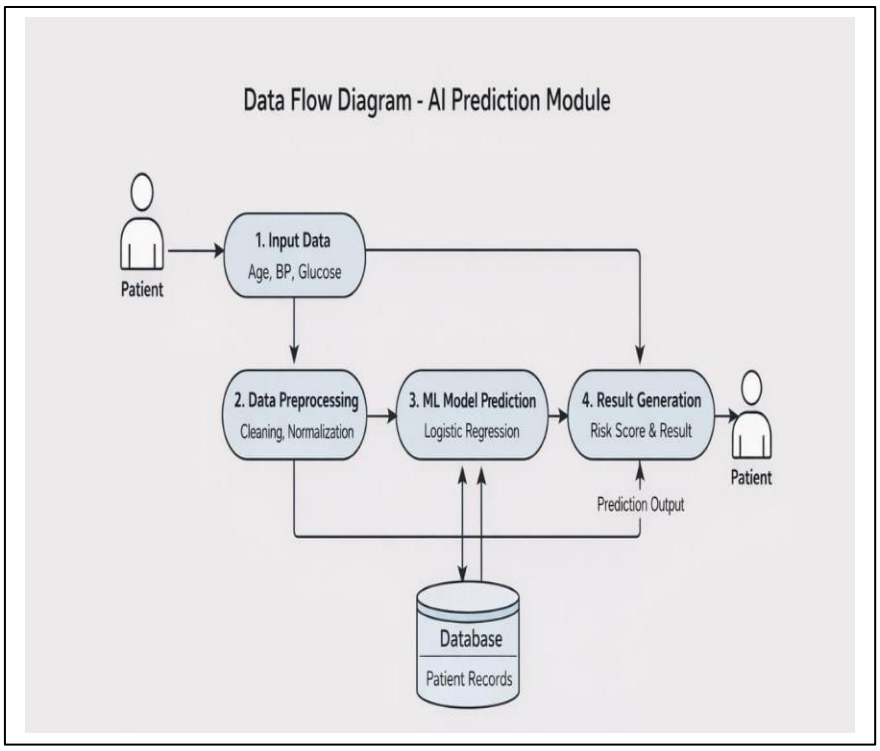


Fig no. 4.4.1 Data-Flow diagram

4.5 UML diagrams set

4.5.1 Use case diagram

The use case diagram shows the main interactions between actors such as Patient, Doctor, and Admin with the system. It illustrates use cases like registration, login, submit health data, view prediction, verify report, manage users, and conduct consultations.

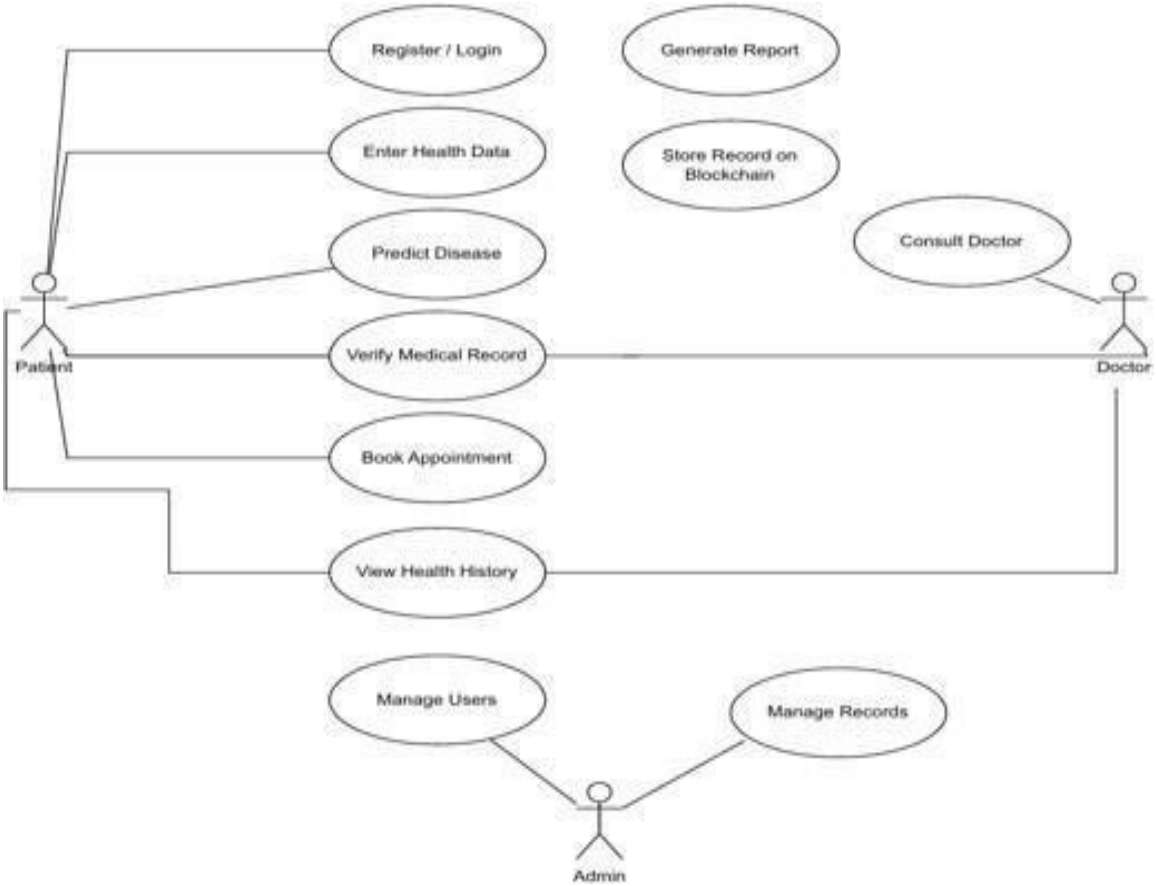


Fig no. 4.5.1.1 Use case diagram

4.5.2 Class diagram

The class diagram represents the main classes, such as User, Patient, Doctor, Prediction, LedgerEntry, and Appointment, along with their attributes and methods. It shows relationships like inheritance between user types and associations between predictions and ledger entries, capturing the static structure of the system.

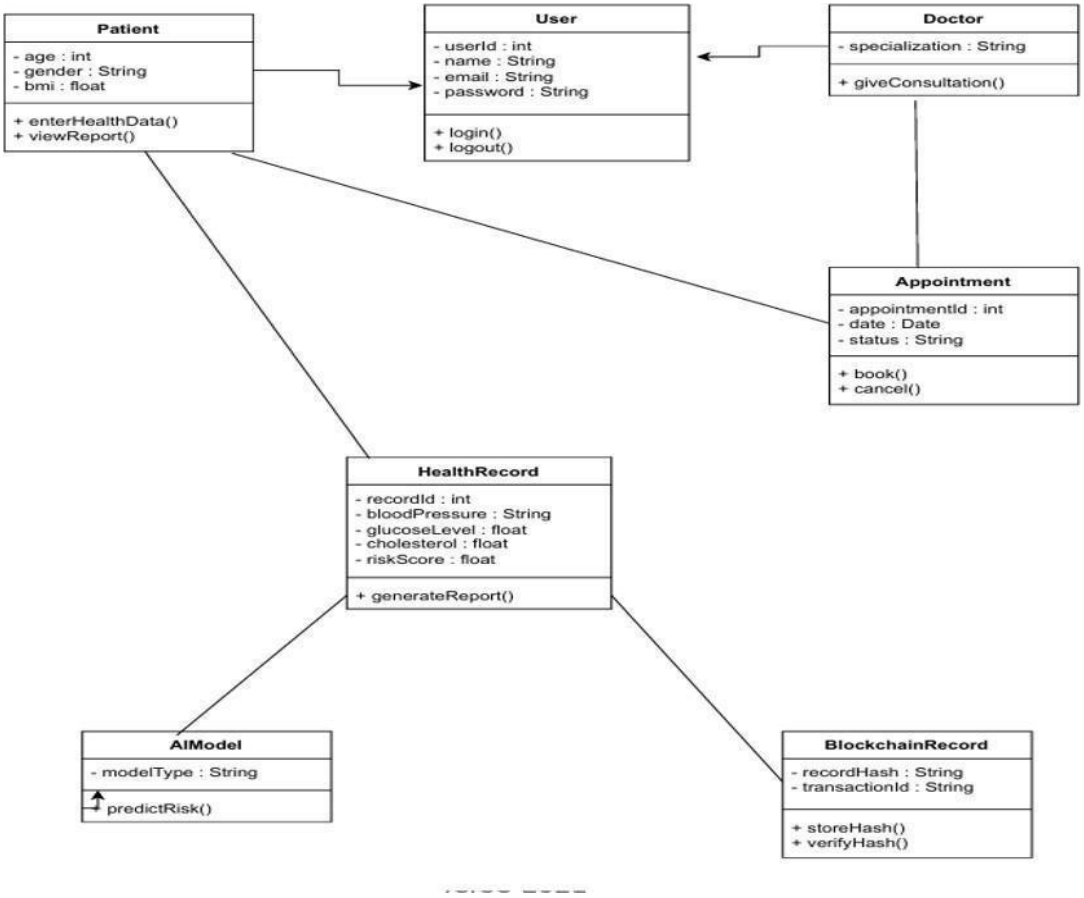


Fig no. 4.5.2.1 Class Diagram

4.5.3 Sequence diagram

The sequence diagram models the order of messages exchanged when a patient submits health data and receives a prediction. It shows interactions between the Patient interface, Backend Controller, AI Service, Database, and Ledger module from request to response.

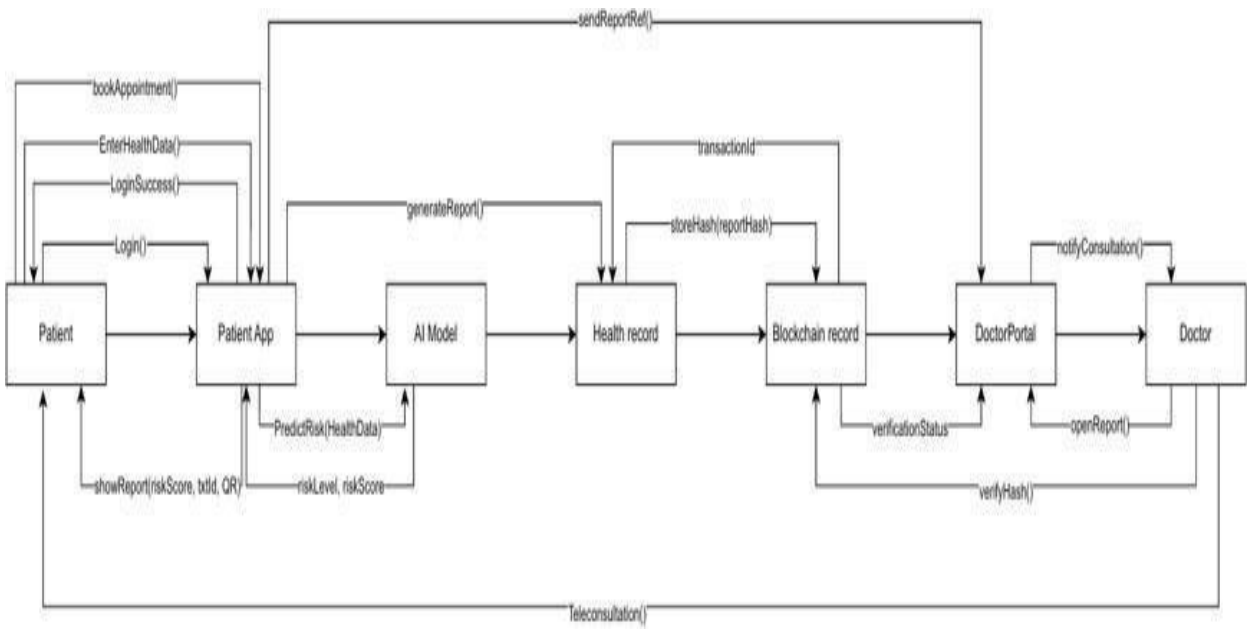


Fig no. 4.5.3.1 Sequence diagram

4.5.4 Activity diagram

The activity diagram captures the workflow of heart disease prediction, from user login and data entry to displaying risk results and storing the hash. It highlights decision points such as data validation and conditional flows, like success or error handling.

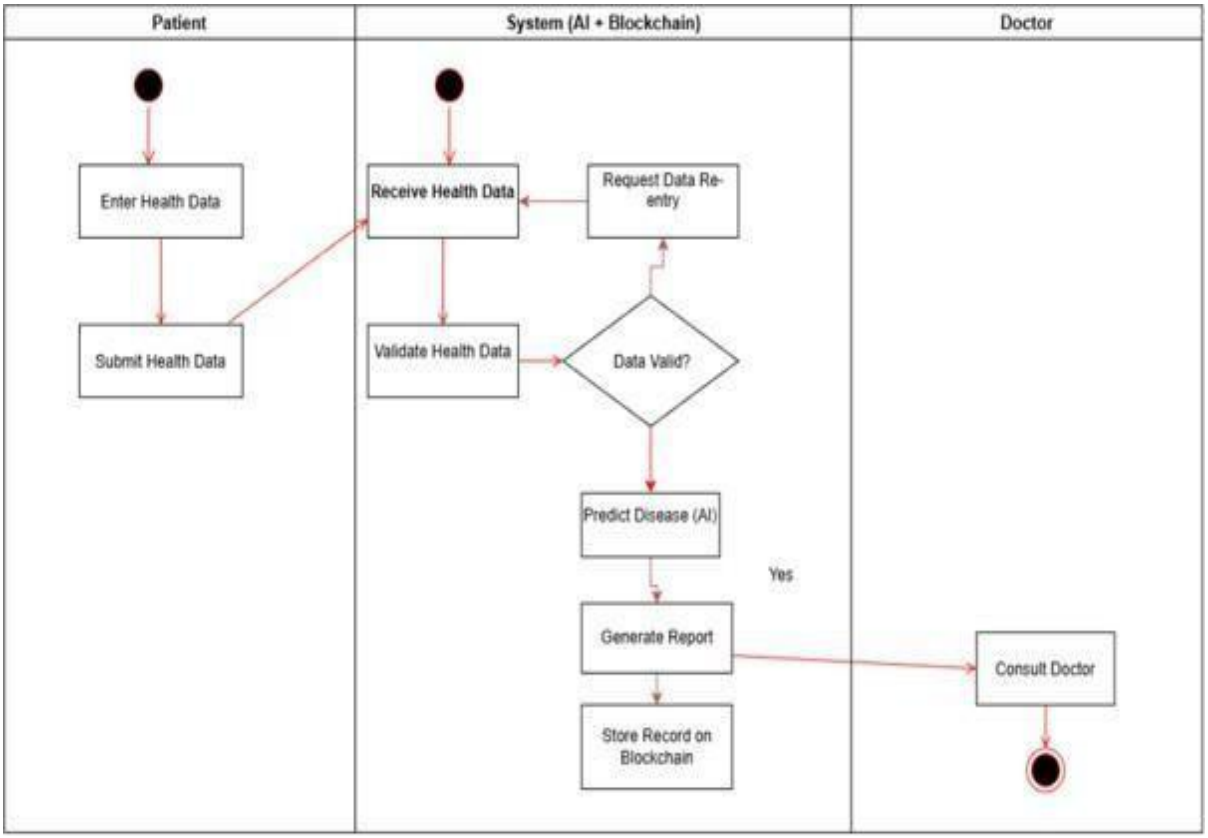


Fig no. 4.5.4.1. Activity diagram

4.5.5 State diagram

The state diagram describes the different states of a prediction record, such as Created, Predicted, Hashed, Verified, and Archived. It shows how transitions occur based on actions like generating prediction, storing a hash, and verifying a report.

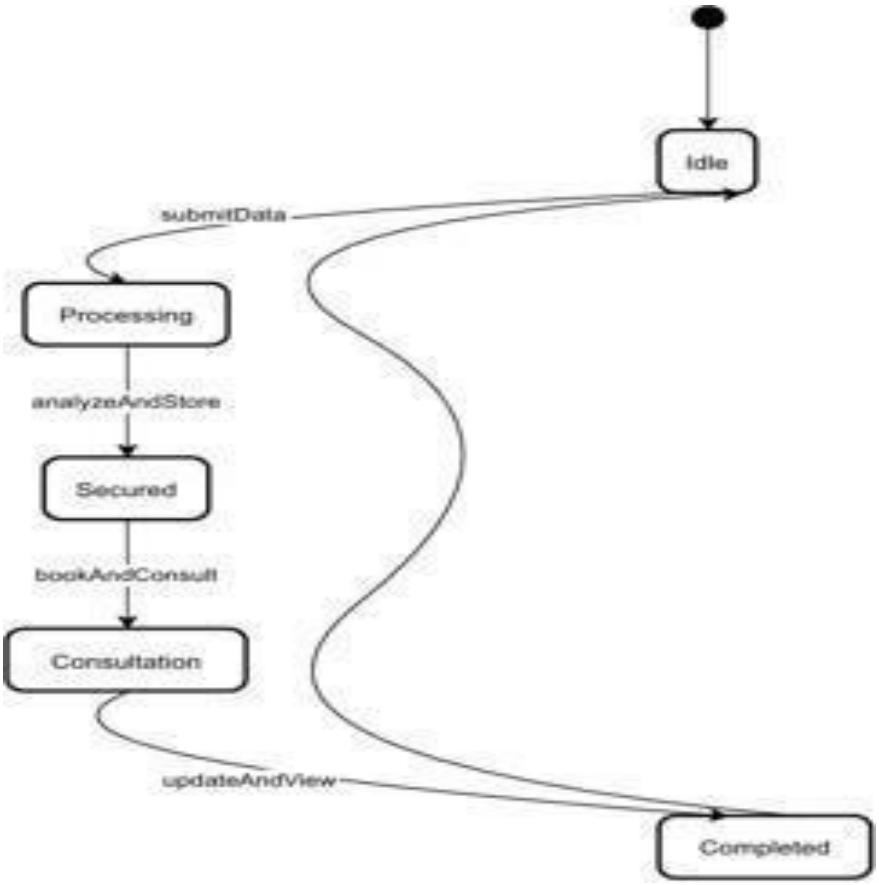


Fig no. 4.5.5.1 State diagram

4.5.6 Component diagram

The component diagram illustrates high-level components such as Web UI, Backend API, AI Model Service, Blockchain/Ledger Service, and Database. It shows how these components are organized and how they depend on each other.

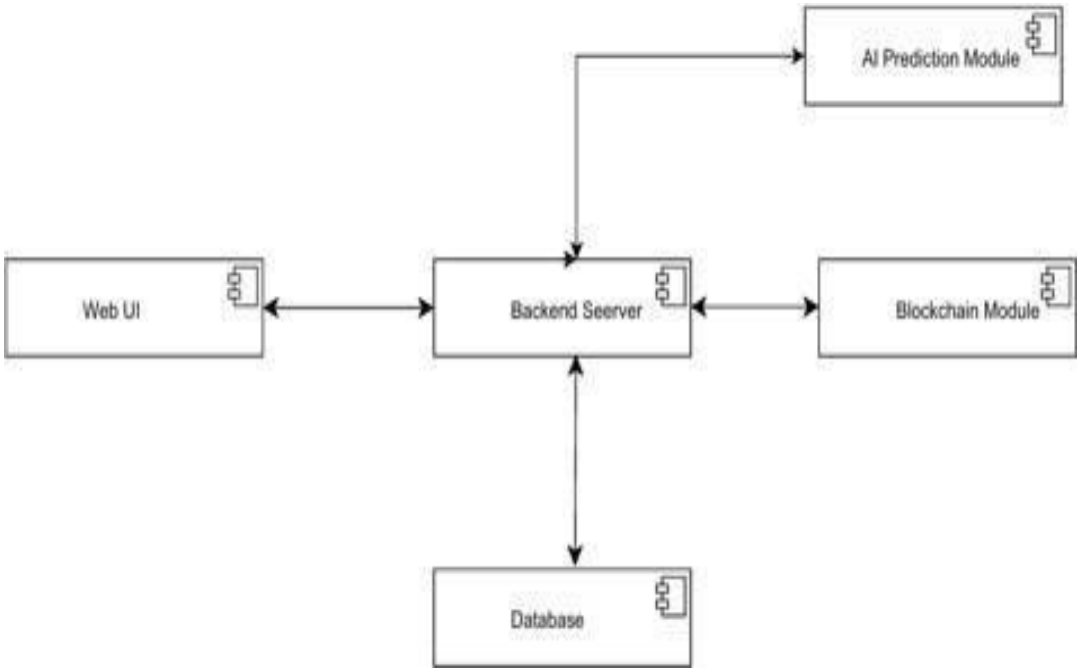


Fig no. 4.5.6.1 Component diagram

4.5.7 Deployment diagram

The deployment diagram represents the physical deployment of software components on nodes such as client browser, application server, and database server. It indicates which components run on which hardware and how they communicate over the network.

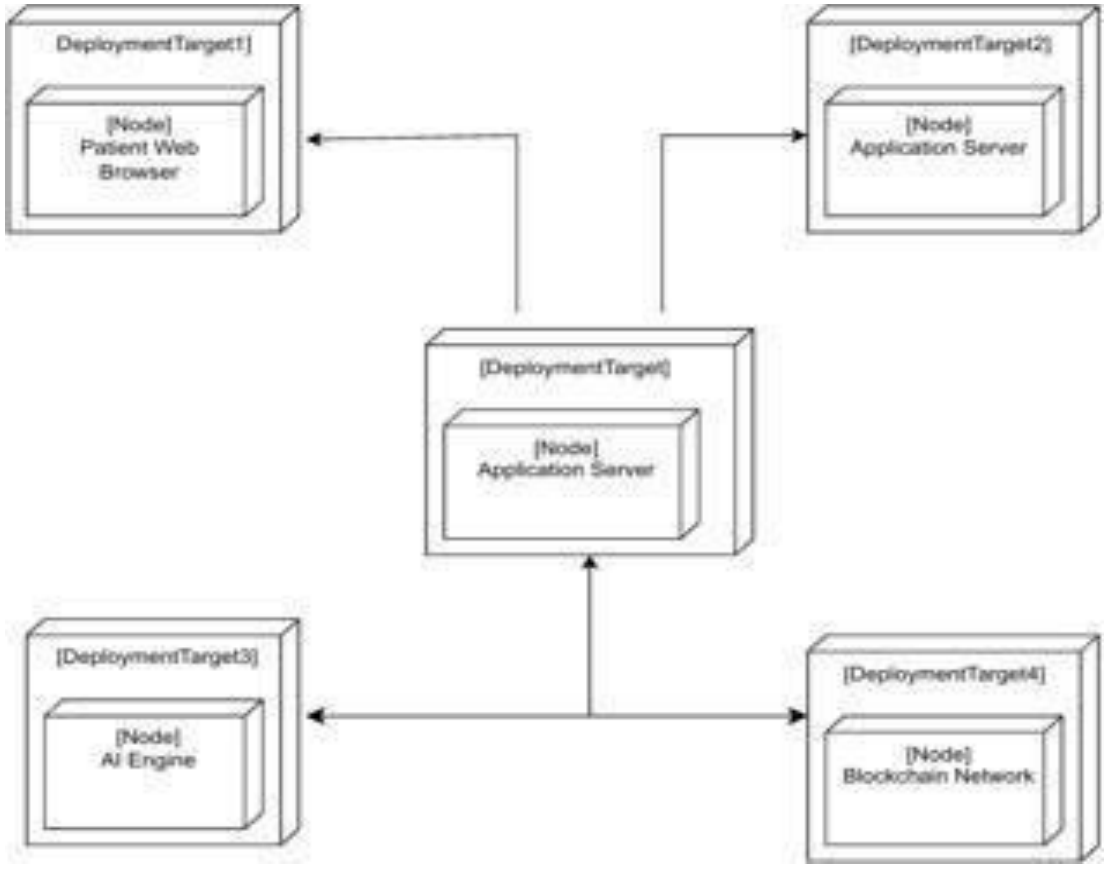


Fig no. 4.5.7.1 Deployment diagram

4.5.8 Package diagram

The package diagram groups related classes and components into logical packages, such as UI, Sevices, Models, Blockchain, and Utilities. It clarifies the modular organization of the codebase.

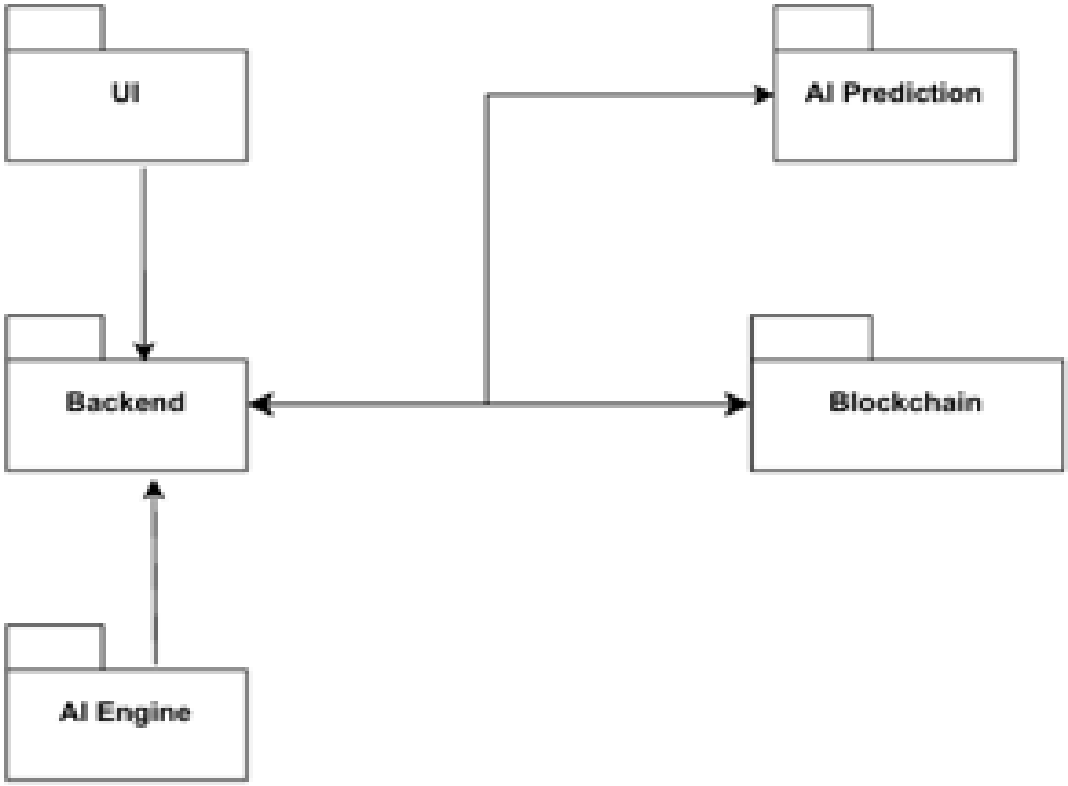


Fig no. 4.5.8.1 Package diagram

4.5.9 Composite structure diagram

The composite structure diagram shows the internal structure of a key class or component, such as the Backend Controller, including its parts and ports. It helps to understand how internal elements collaborate to fulfil system behaviour.

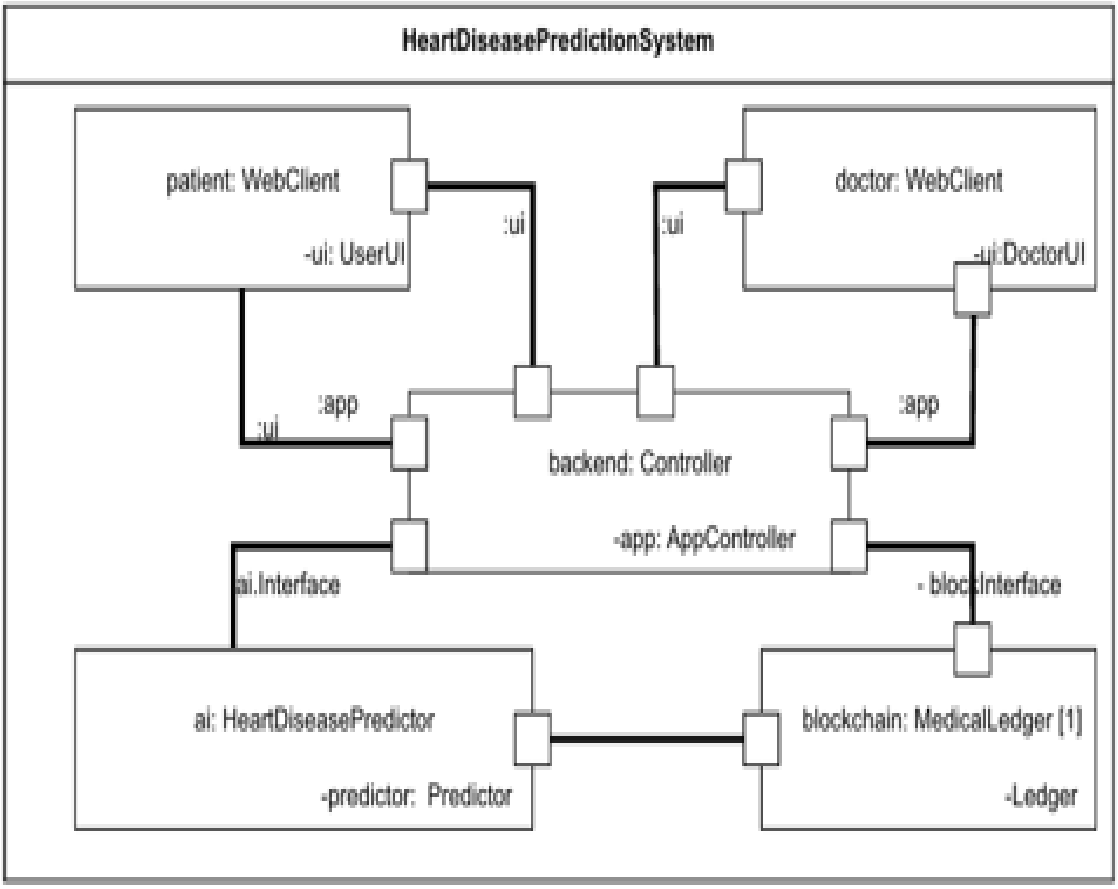


Fig no. 4.5.9.1 Composite Structure diagram

4.5.10 Profile diagram

The profile diagram defines stereotypes and tagged values that customize UML elements for the healthcare and blockchain domains. It can include stereotypes for entities like “HealthcareRecord” or “LedgerTransaction.”

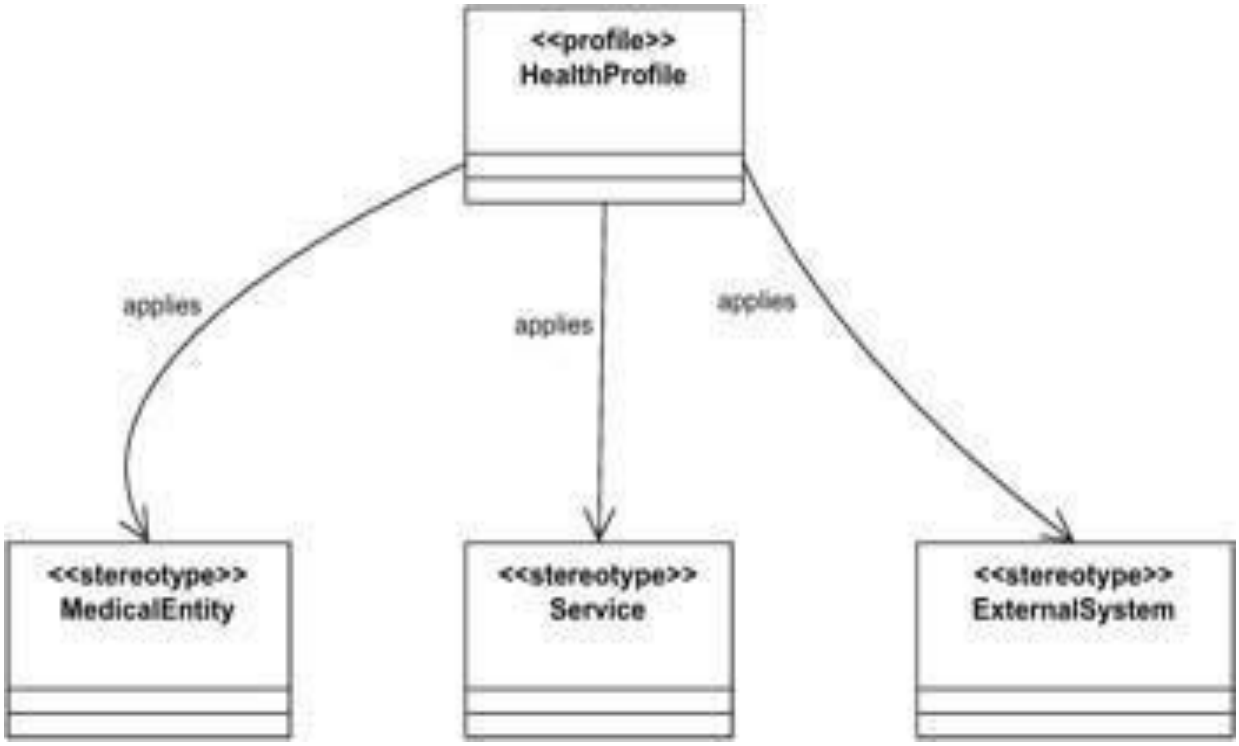


Fig no. 4.5.10.1 Profile diagram

4.5.11 Communication diagram

The communication diagram expresses interactions among objects in terms of links and numbered messages for a typical scenario, like report verification. It complements the sequence diagram by focusing on structural relationships.

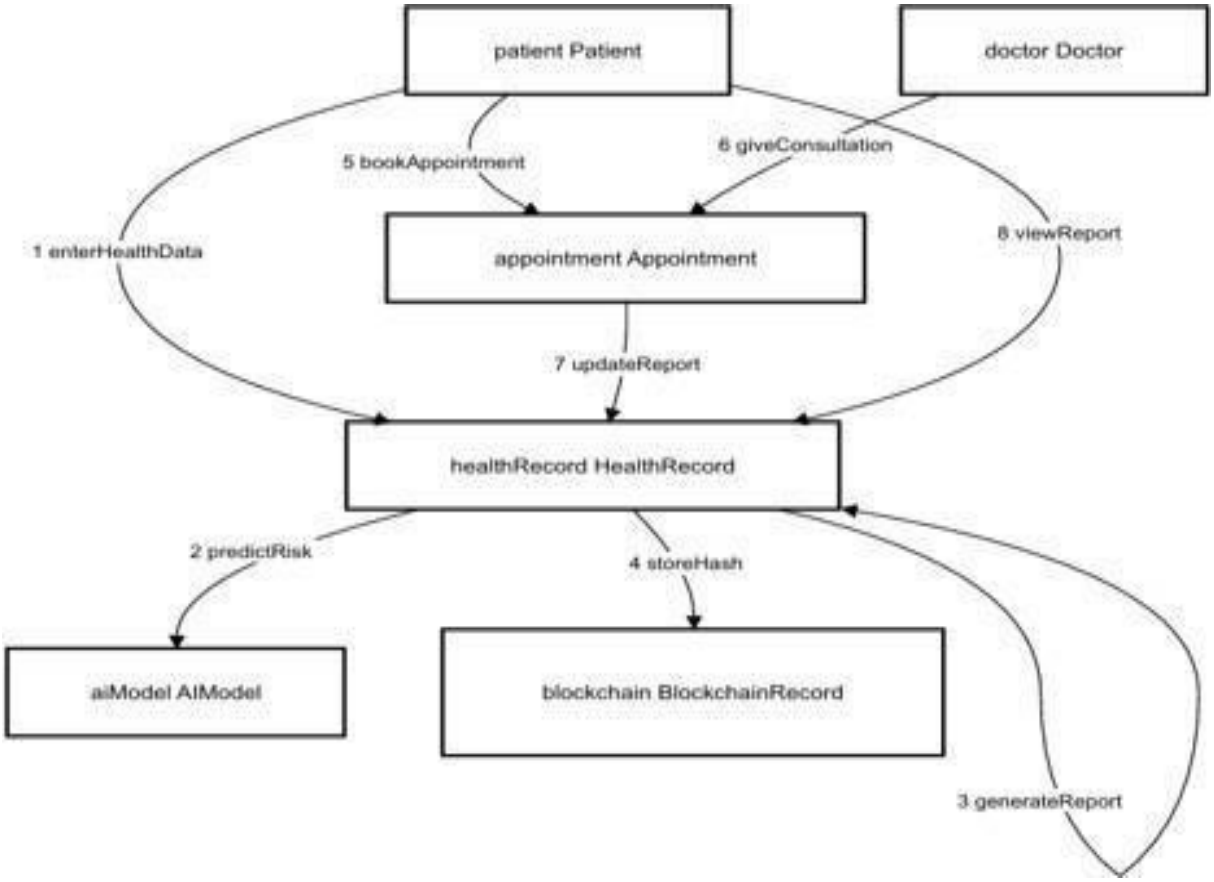


Fig no. 4.5.11.1 Communication diagram

4.5.12 Interaction overview diagram

The interaction overview diagram summarizes multiple interaction diagrams, linking different scenarios such as prediction, telemedicine consultation, and verification. It provides a high-level view of how various interactions connect.

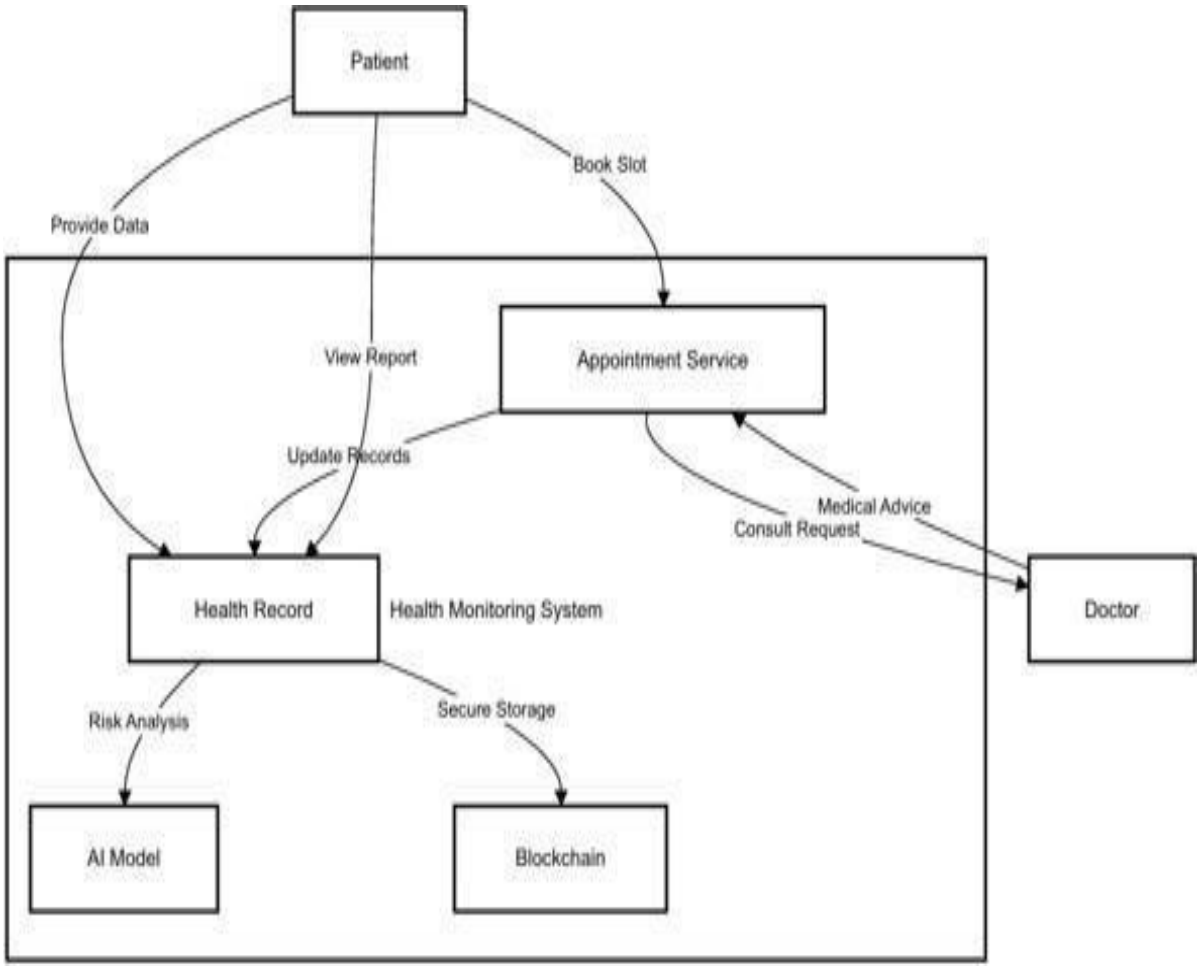


Fig no. 4.5.12.1 Interaction Overview diagram

4.5.13 Timing diagram

The timing diagram shows how the state of key objects, such as the prediction record or user session, changes over time during a scenario. It highlights timing constraints and the ordering of events.

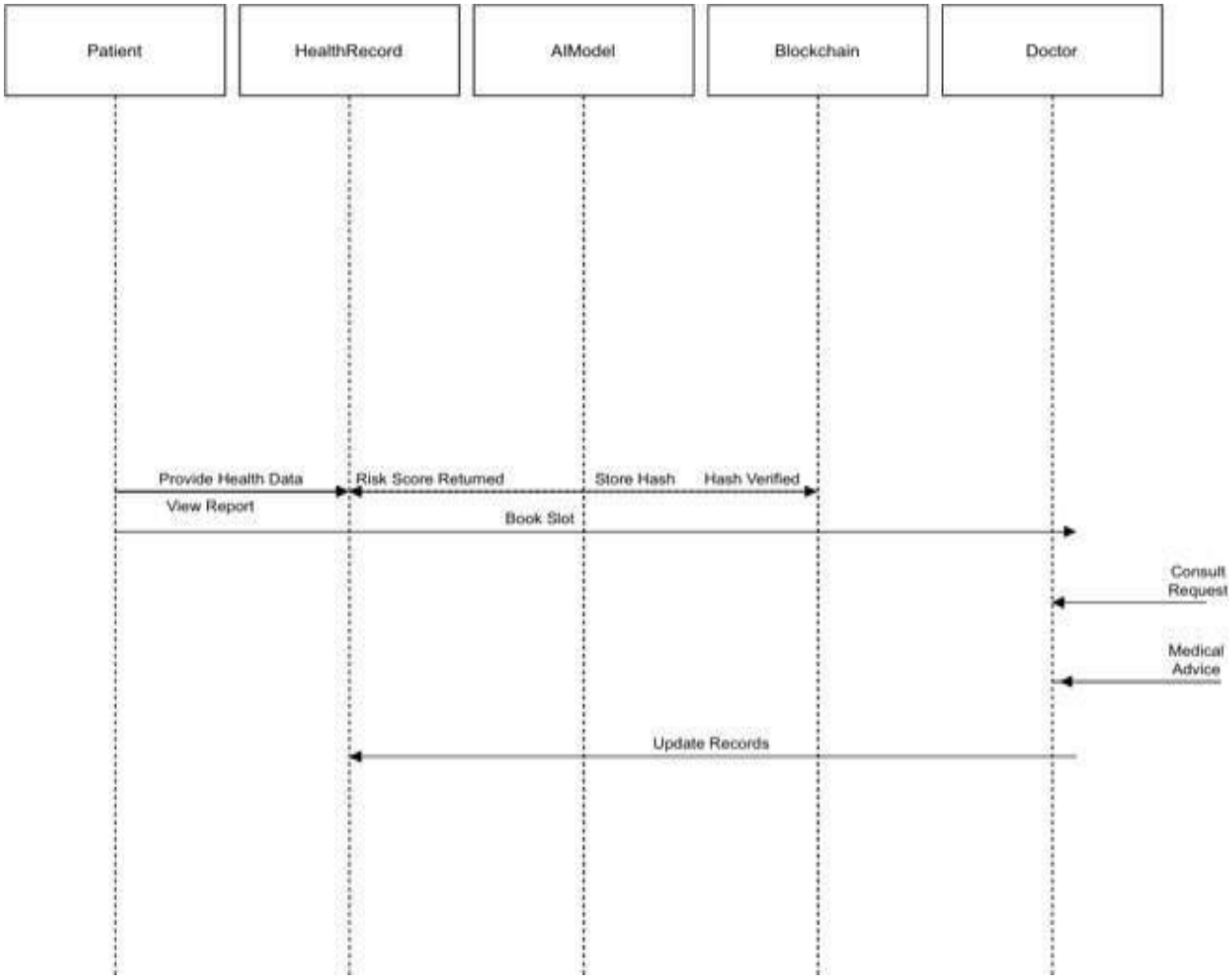


Fig no. 4.5.13.1 Timing diagram

4.5.14 Additional UML diagram

An additional UML diagram, such as another detailed sequence or activity diagram for telemedicine booking, can be used to capture specific behaviour. It illustrates the steps involved when patients schedule and attend an online consultation.

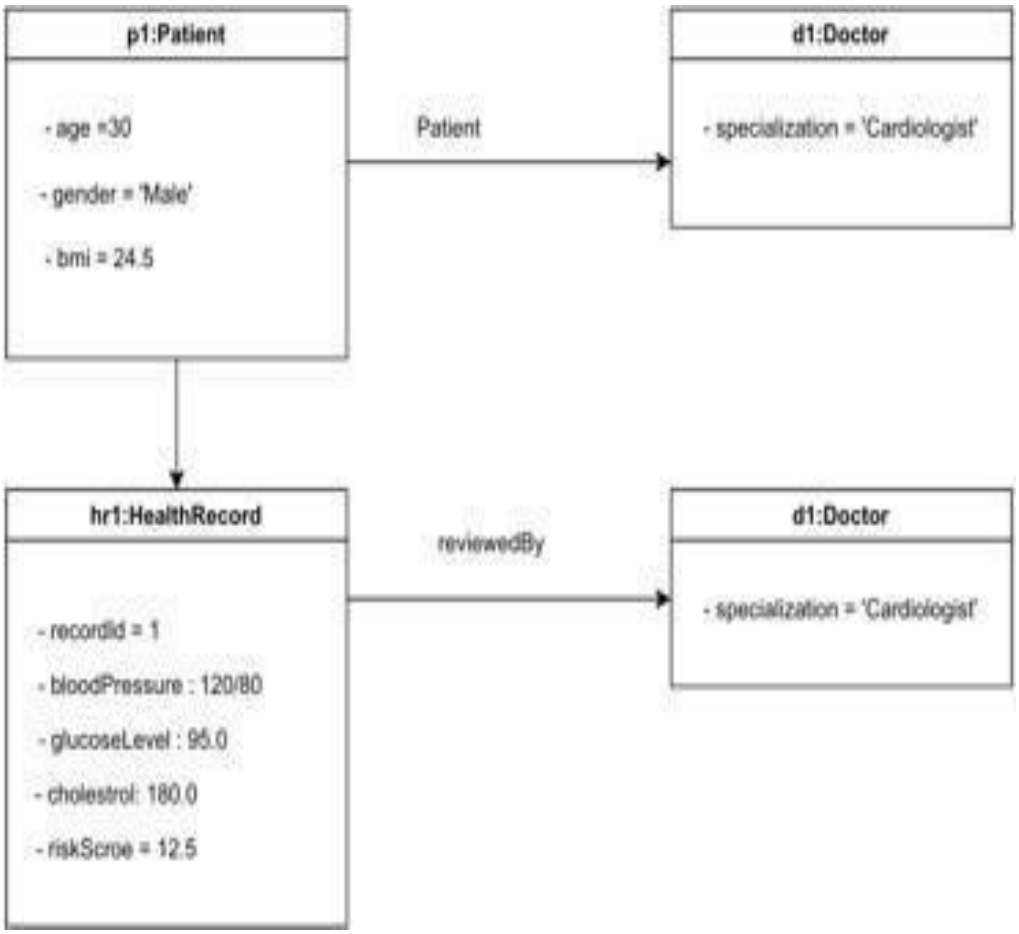


Fig no. 4.5.14.1 Additional UML diagram

4.6 Design decisions and constraints

The system adopts a web-based architecture to maximize accessibility from different devices and locations. A Decision Tree model is chosen for prediction because it is interpretable and easy to explain to medical practitioners. A private permissioned blockchain implemented as a ledger table with hash chaining is used rather than a public blockchain to maintain confidentiality and simplify deployment. Constraints include limited dataset size, development time, and the need to simulate blockchain logic within the available infrastructure.

CHAPTER-5

SYSTEM IMPLEMENTATION

5. System Implementation

5.1 Implementation Environment

Programming Languages

- Python 3.x – Used for backend development and AI/ML model implementation.
- TypeScript – Used for frontend application logic in React.
- HTML & CSS – Used for designing user interface components.
- SQL – Used for database operations and query execution.

Frameworks / Libraries

- Flask – Backend web framework used for REST APIs, authentication, and server logic.
- React.js – Frontend framework used to build interactive web interfaces.
- Tailwind CSS – Used for responsive UI styling.
- Scikit-learn – Used to implement the Decision Tree prediction model.
- Pandas – Used for dataset handling and preprocessing.
- NumPy – Used for numerical and mathematical computations.
- SHA-256 Hashing – Used for blockchain-style ledger security and report integrity.

IDE / Tools Used

- Visual Studio Code (VS Code) – Used for coding and debugging.
- Postman – Used for API testing and verification.
- GitHub – Used for version control and project management.
- Terminal / Command Prompt – Used to run backend and frontend servers.

Operating System

- Windows / Linux – Used as development and deployment environments.

Database / Server Details

- Database: PostgreSQL / Supabase
- Server: Flask Local Server (REST APIs)
- Storage: Patient records, predictions, and blockchain ledger logs

5.2 Module-wise Implementation

5.2.1 User Authentication Module

Purpose

The User Authentication Module provides secure access to the Smart Healthcare System for registered users such as patients and doctors. It ensures that only authorized users can access the system and interact with prediction reports and healthcare features.

Algorithms / Logic Used

• User registration and login data are collected through secure web forms.

• Input validation ensures required fields are filled and the data format is correct.

• User credentials are securely stored in the system database.

• Role-based access control allows login for patients and doctors.

• After successful login, the system redirects the user to the appropriate dashboard.

Input

• Username / Email

• Password

• User role (Patient / Doctor)

• Basic profile information

Output

• Successful login authentication

• Access to the user dashboardError message for invalid credentials

5.2.2 Data Processing Module

Purpose

The Data Processing Module is responsible for collecting and preparing patient health data before it is used by the machine learning model for prediction.

Functions / Processing Logic

• The system uses the Heart Disease dataset obtained from Kaggle for training the prediction model.

• The dataset contains multiple medical attributes such as age, cholesterol level, blood pressure, heart rate, and other clinical parameters.

• Data preprocessing is performed to remove missing values and organize the dataset into a structured format.

• Patient input values are validated and converted into the format required by the prediction model.

Input

• Patient health parameters entered through the system interface.

• Kaggle heart disease dataset used for model training.

Output

• Clean and structured data ready for prediction.

5.2.3 Prediction / Analysis Module

Purpose

The Prediction Module analyzes patient health parameters and predicts the risk of heart disease using a machine learning model.

Tools / Technologies Used

• Machine Learning model trained using the Kaggle Heart Disease dataset.

• Decision Tree classification algorithm.

Processing Logic

• Load the trained machine learning model.

• Receive processed health data from the data processing module.

• Apply the Decision Tree model to classify whether the patient has heart disease risk.

• Generate prediction results with risk indication.

Input

• Preprocessed patient health parameters.

Output

• Prediction result (Heart Disease/ No Heart Disease).

• Risk level classification.

5.2.4 Blockchain / Ledger Module

Purpose

The Blockchain Module ensures secure storage of prediction reports so that medical data cannot be modified or tampered with.

Processing Logic

• Generate a cryptographic SHA-256 hash of the prediction report.

• Store the hash in a blockchain-style ledger table.

• Link each hash with the previous ledger entry to maintain data integrity.

• Generate a QR code or transaction ID for report verification.

Input

• Prediction result

• Patient ID

• Timestamp

Output

• Blockchain hash entry

• QR code or transaction ID for report verification

5.2.5 Telemedicine Module

Purpose

The Telemedicine Module enables communication between patients and doctors through the platform, allowing doctors to review prediction reports and provide medical consultation based on the results.

Functions / Processing Logic

• Allows doctors to access patient prediction reports generated by the system.

• Enables patients to share their reports with doctors for medical consultation.

• Allows doctors to review patient health data and provide recommendations or consultation notes.

• Updates patient records after consultation for future reference.

Input

• Patient prediction report

• Patient health data

• Doctor consultation request or response

Output

• Doctor recommendations or consultation notes

• Updated patient records

• Communication history between patient and doctor

5.3 Algorithm / Pseudo Code

The main algorithm is the Decision Tree-based heart disease prediction flow. At a high level, the algorithm loads the dataset, preprocesses features, trains the Decision Tree classifier, and then uses it in prediction.

- Step 1: Start the Smart Healthcare System.

  Step 2: User logs in using valid credentials.

  Step 3: The system redirects the user to the dashboard.

  Step 4: Patient enters health parameters through the interface.

  Step 5: The system validates the input data.

  Step 6: The Data Processing Module prepares the input data.

  Step 7: Load the trained Decision Tree model built using the Kaggle dataset. Step 8: Apply the machine learning model to predict heart disease risk.

  Step 9: Generate prediction result.

  Step 10: Generate a SHA-256 hash of the prediction report.

  Step 11: Store hash in blockchain-style ledger table.

  Step 12: Generate QR code or transaction ID. Step 13: Display prediction results to the patient.

  Step 14: Doctor can view the report and provide consultation.

  Step 15: End the process.

Flow of Execution:

Login

↓

User Role Selection (Patient / Doctor)

↓

Patient Inputs Health Parameters

↓

Data Validation

↓

AI Prediction Model (trained using Kaggle dataset)

↓

Prediction Result Generation

↓

Blockchain Hash Generation

↓

Ledger Storage

↓

QR Code / Transaction ID Generation

↓

Report Display and Doctor Consultation

5.4 Screenshots and Explanation

- **Login Page:** The login screen allows patients, doctors, and admins to enter their credentials and access role-specific dashboards. It validates inputs and redirects users upon successful authentication.

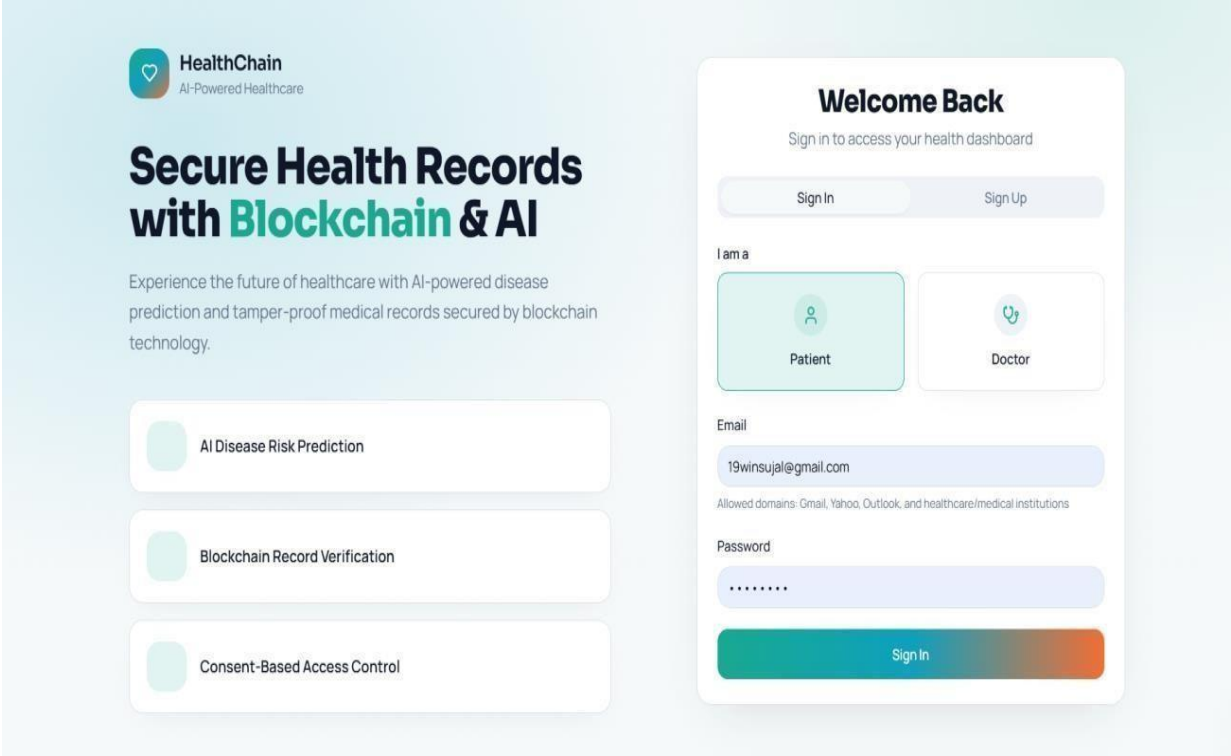


Fig no. 5.4.1 Patient/Doctor login Page

- **Prediction Result Page:** This screen shows the output of the AI model, including disease/no disease status, risk category, and a summary of submitted parameters. It also provides a QR code or transaction ID for verification.

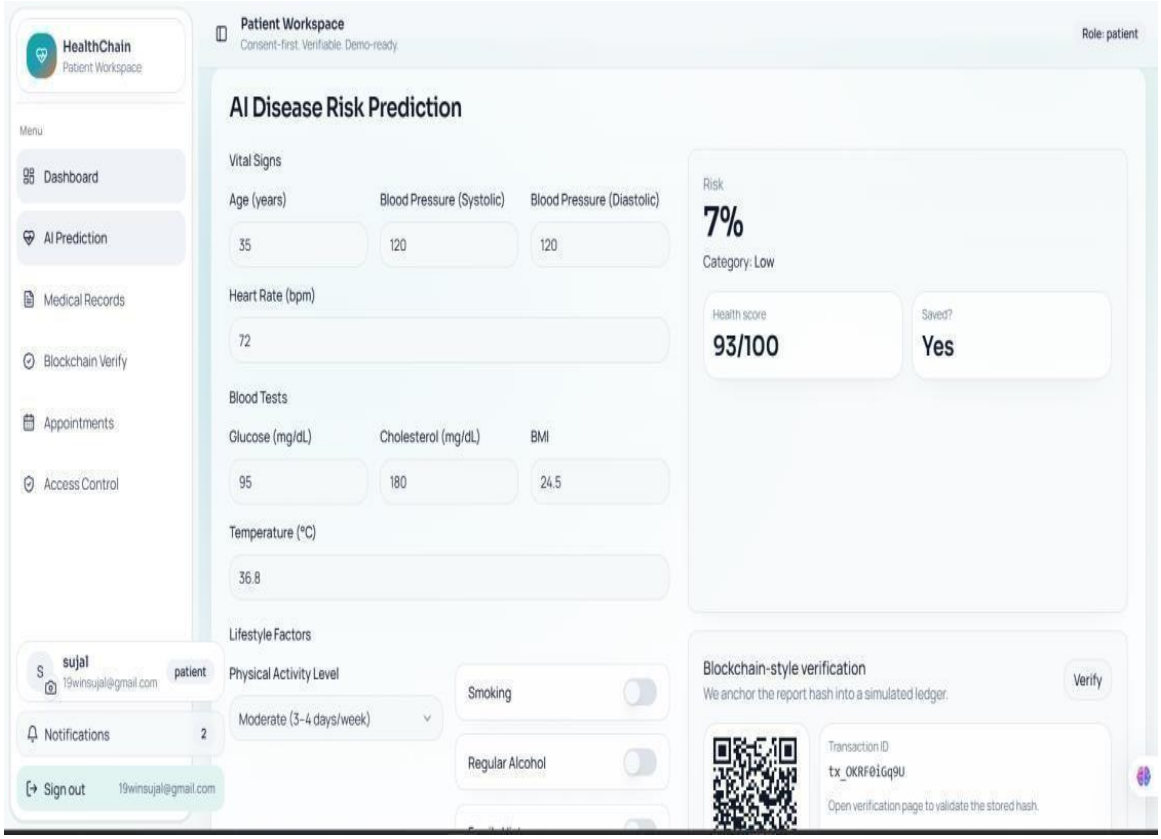
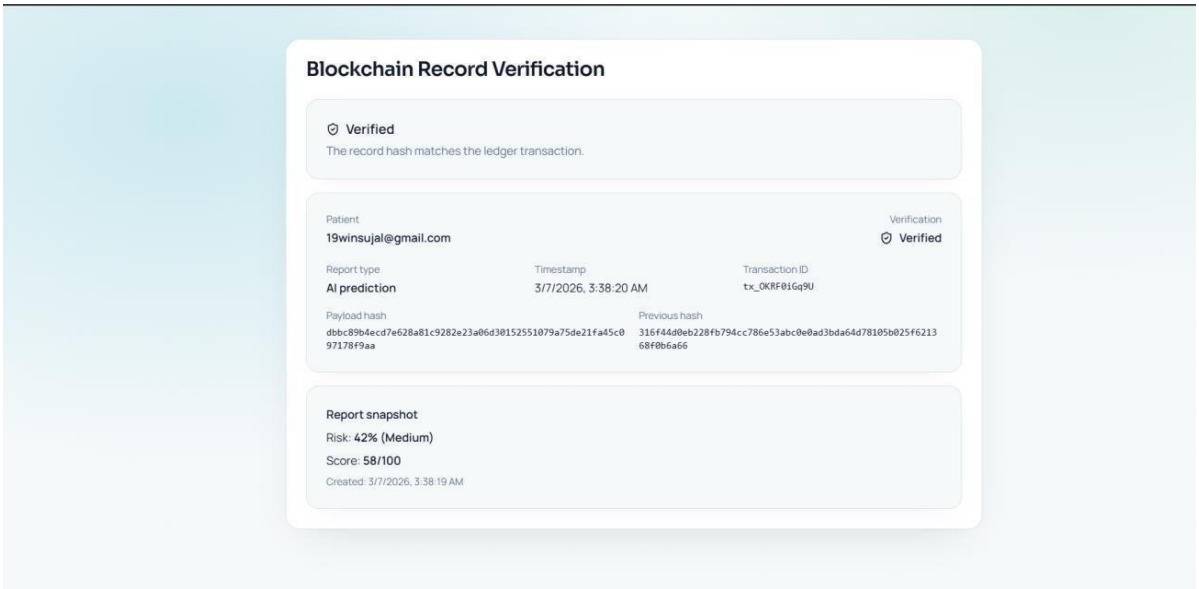


Fig no. 5.4.2 AI Disease Prediction

- **Verification Page:** The verification page lets users scan a QR code or enter a transaction ID to confirm whether a report is authentic by comparing hash values in the ledger.



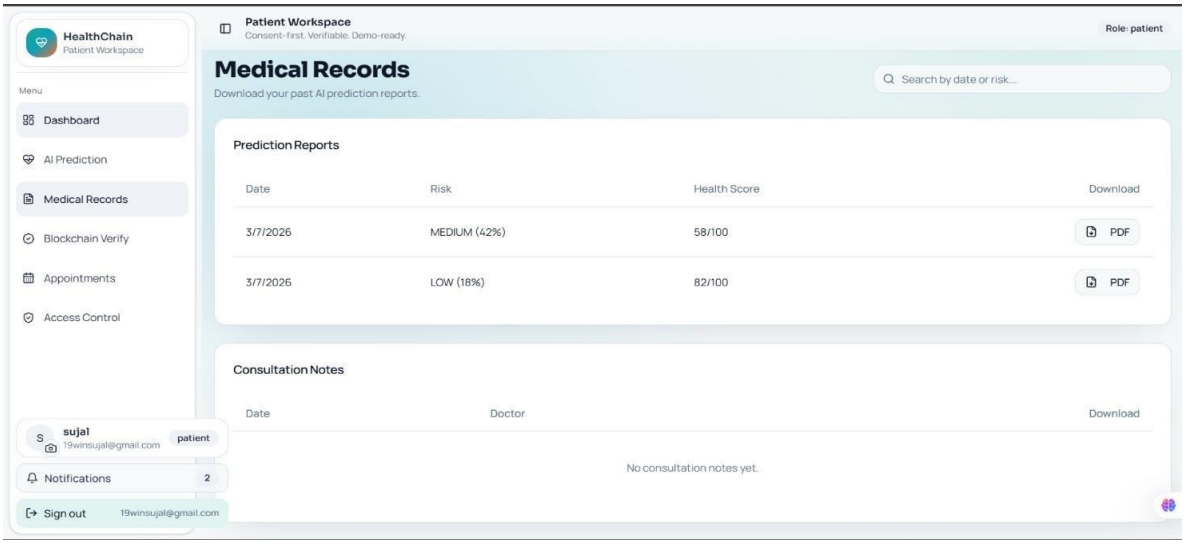


Fig no. 5.4.3 Blockchain record verification

- **Medical Records:** This page displays a patient's medical records, including AI-generated prediction reports with risk levels, health scores, and downloadable PDFs.It also provides a section for consultation notes and allows easy navigation to other healthcare features like appointments and blockchain verification.

- **Appointment page:** The appointments page allows users to book consultations by selecting a date, entering a doctor ID, and providing a reason for the visit. It also displays scheduled appointments, chat status, and doctor notes, enabling users to track and manage their consultations efficiently.

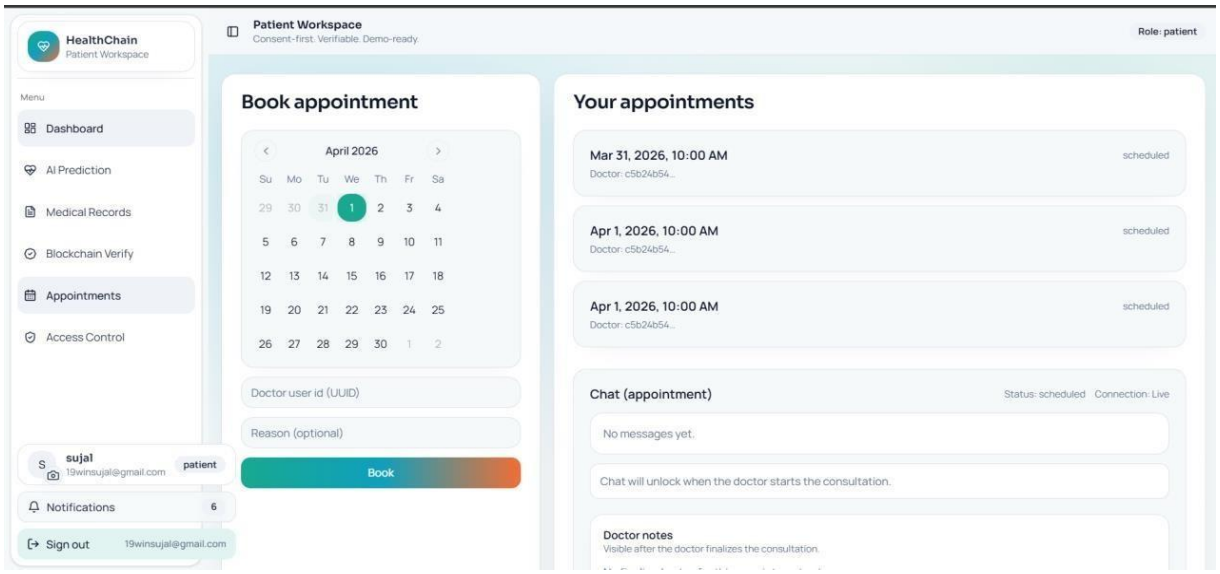
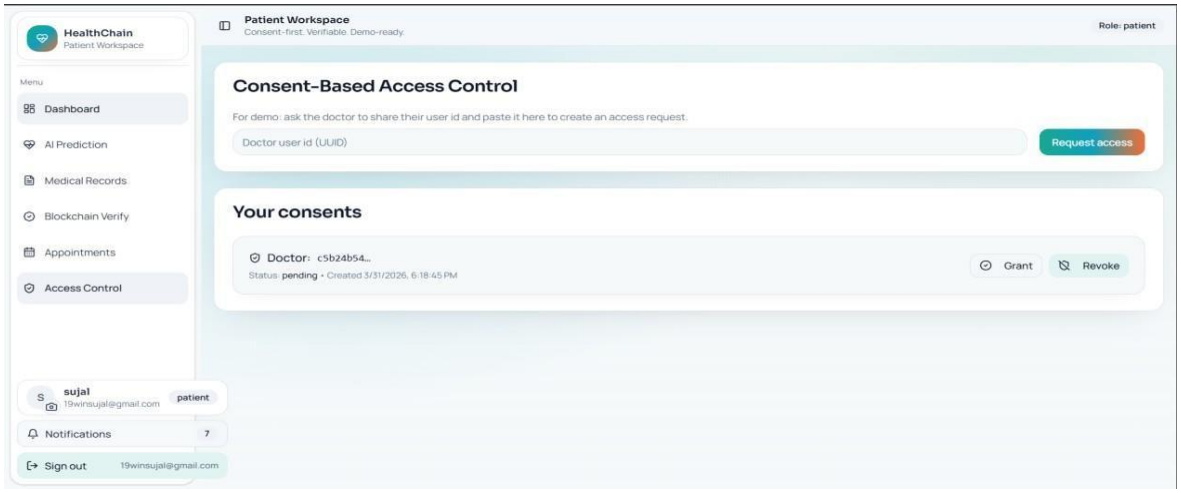


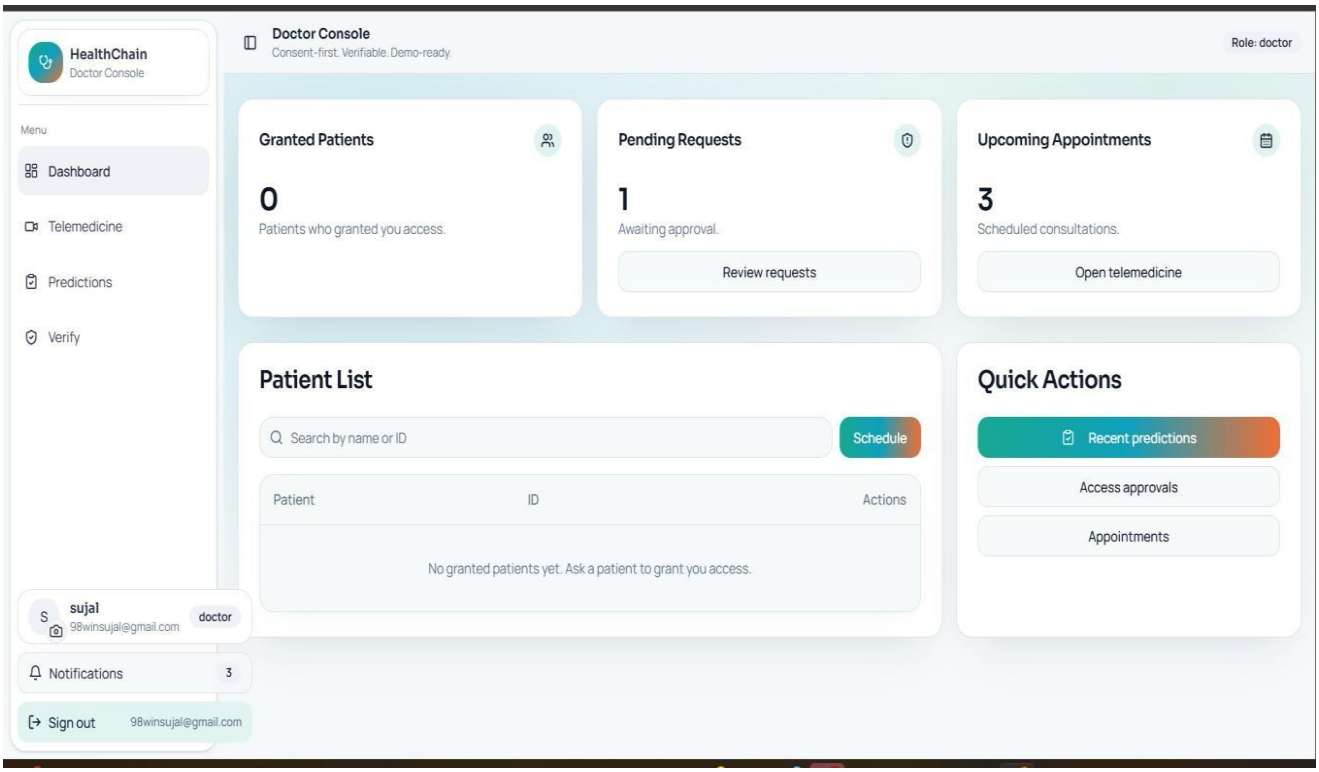
Fig no. 5.4.4 Appointment booking

- **Consent-Based Access Control:** The access control page lets users request, grant, or revoke access to their medical data using doctor IDs. It ensures secure and consent-based sharing of health information.



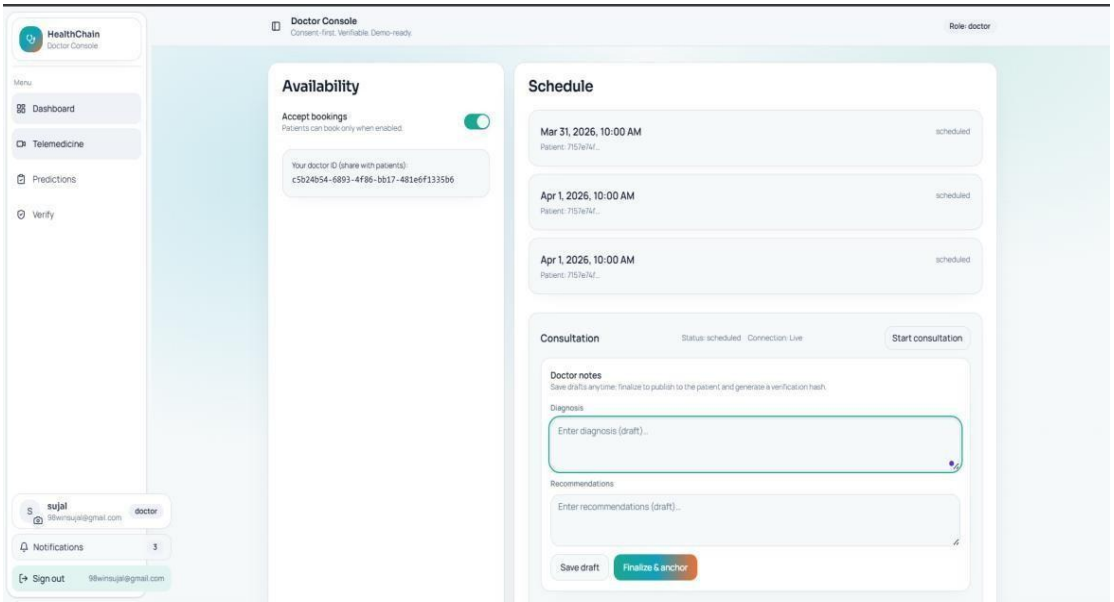
- **Doctor Dashboard Page:** The doctor dashboard lets users manage patients, access requests, and upcoming appointments in one place.

  It also provides quick actions for predictions, approvals, and consultation management.

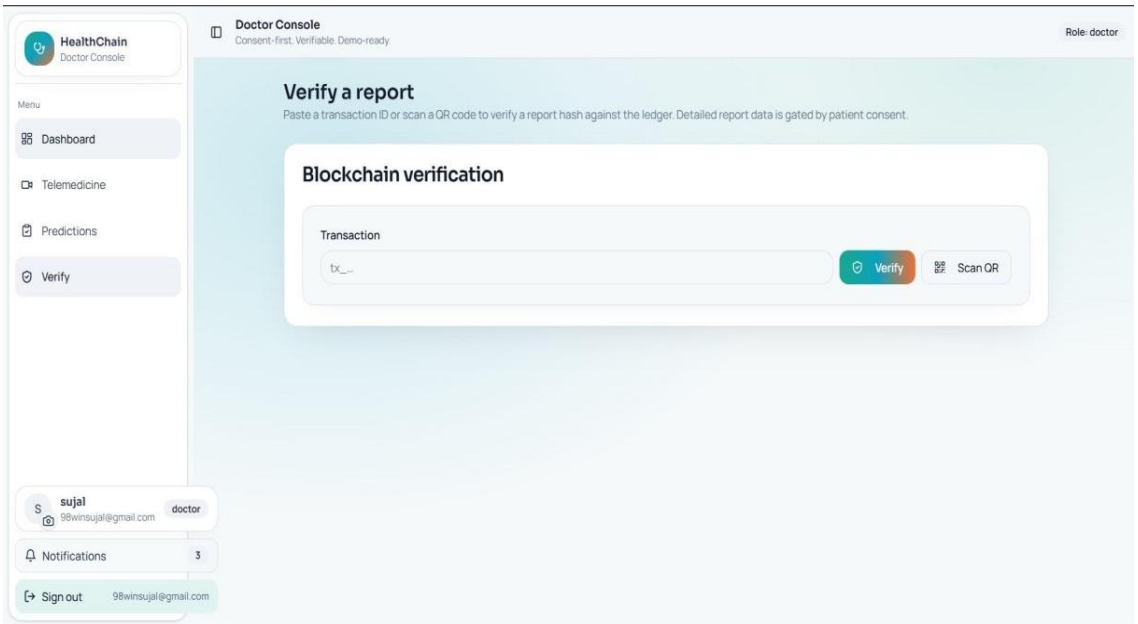


- **Telemedicine Page:** The telemedicine page lets doctors manage availability, view

  schedules, and conduct consultations. It also allows adding and finalizing doctor notes, diagnoses, and recommendations.



- **Verification Page:** The verification page lets users validate reports using a transaction ID or QR code. It ensures authenticity through secure blockchain-based verification.



CHAPTER-6

TESTING & VALIDATION

6 Testing and Validation

Testing ensures that the Smart Healthcare System functions correctly, predicts heart disease accurately, and securely stores medical records using blockchain verification. The platform was tested at multiple levels to validate prediction accuracy, module integration, and overall system reliability.

6.1 Testing Strategy

The testing strategy followed a structured approach consisting of:

- Unit Testing
- Integration Testing
- System Testing

Each level verifies different aspects of the system.

6.1.1 Unit Testing

Objective

To verify the correctness of individual modules independently

Modules Tested

1. User Authentication Module
2. Data Processing Module
3. Prediction (Decision Tree) Module
4. Blockchain / Ledger Module
5. Telemedicine Module
6. Database Management Module

Testing-Process

Result

- Login tested with valid credentials.
- Login tested with invalid credentials
- Patient data preprocessing tested with complete and missing values.
- Decision Tree model tested with sample records for correct risk prediction.
- SHA-256 hash generation tested for consistency and uniqueness.
- Ledger entries tested for proper hash chaining.
- Telemedicine features tested for report viewing and consultation updates.
- Database queries tested for correct insertion and retrieval of records.All modules produced expected outputs under normal and edge conditions.

6.1.2 Integration Testing

Objective

To ensure that different modules interact correctly when combined.

Integration Scenarios Tested

1. Patient data submission → Data preprocessing → AI prediction → Database insertion
2. Prediction result → SHA-256 hash generation → Blockchain ledger storage
3. Ledger entry → QR code generation → Report verification
4. Doctor login → Report access → Telemedicine consultation

Testing-Process

Result

- Verified that patient records and prediction results were stored correctly in the database.
- Confirmed that SHA-256 hashes were generated and linked properly in the ledger.
- Ensured that reports were correctly displayed on patient and doctor dashboards.
- Checked that role-based access control restricted unauthorized actions.
- Validated that verification using QR code or transaction ID returned correct report integrity.All interconnected modules functioned correctly without data loss or mismatch.

6.1.3 System Testing

Objective

To validate the complete system performance and functionality under real-world usage conditions.

System Scenarios Tested

1. Multiple patients entering health data simultaneously
2. Consecutive AI predictions for different patient records
3. Prediction result storage with blockchain hash chaining
4. Doctors accessing reports and consultation features
5. End-to-end workflow from data input to report verification

Testing Process

- Verified that the entire workflow executed without interruption.

Result

Checked that predictions were generated accurately and displayed correctly.

- Confirmed that blockchain hashes were created and stored for every report.
- Ensured reports were retrievable through patient and doctor dashboards.
- Tested role-based access for patients, doctors, and admins.
- Validated QR code and transaction ID verification functionality.
- Monitored system response time under multiple user requests.
- The complete system operated successfully with stable performance, secure data handling, and accurate predictions.

6.2 Test Cases

Example table format:

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC01	Valid patient login	Correct username and password	User logged in and redirected to the appropriate dashboard	As expected	Pass
TC02	Heart disease prediction with valid data	Complete set of patient parameters	Prediction result displayed with risk level	As expected	Pass
TC03	Report hash verification	Valid transaction ID or QR scan	Integrity confirmed and report displayed	As expected	Pass
TC04	Missing	Incomplete	Error message and	As	Pass

Table no. 6.2.1 Test case for Functionality of the System

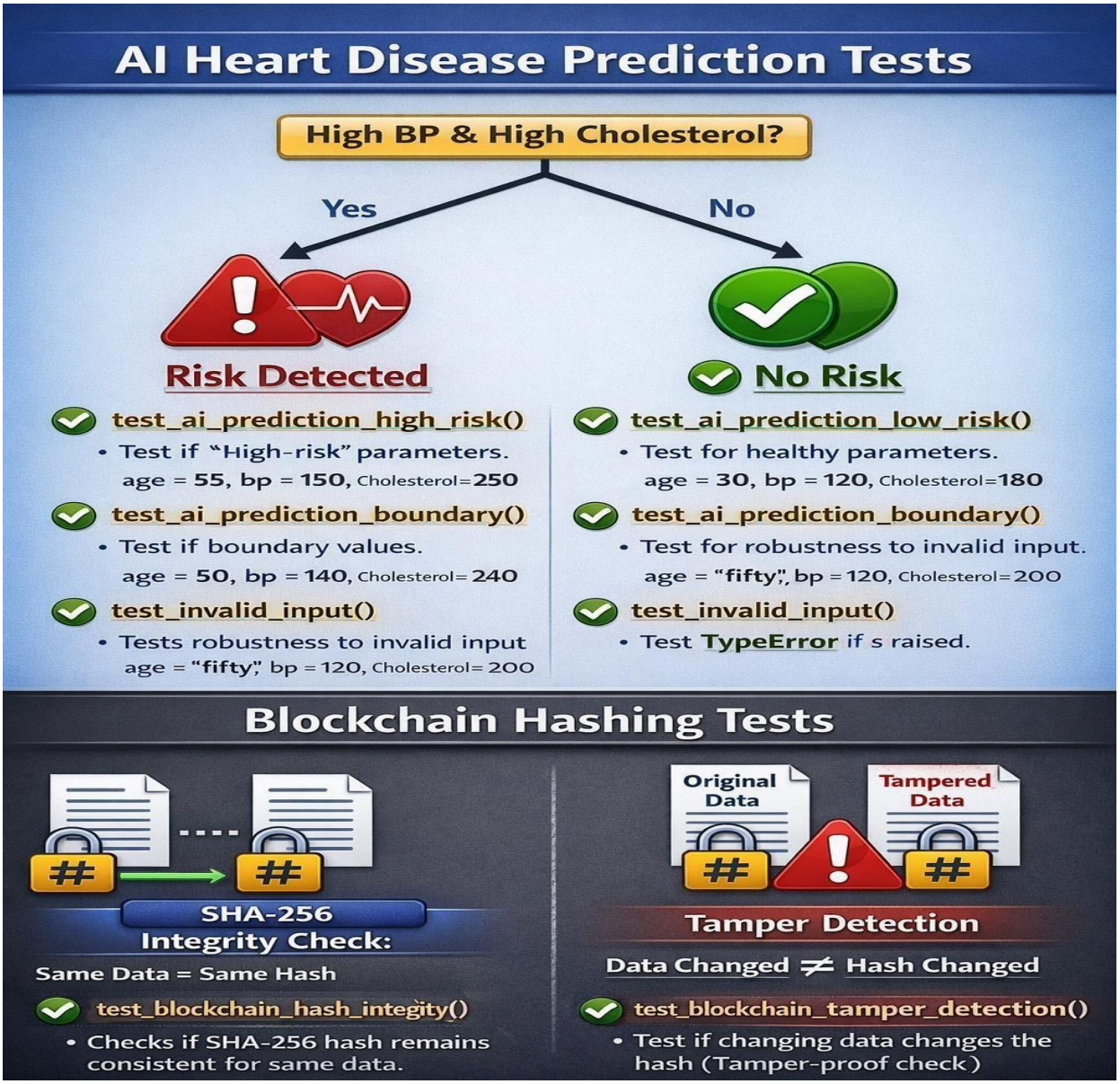


Fig no. 6.2.2 Prediction & Hashing Tests

6.3 Test Results

Summary of Passed/Failed Cases

- Total Test Cases Executed: 6
- Passed: 6
- Failed: 0
- Critical Failures: 0

All major functionalities of the system performed as expected during testing.

Observations

- AI predictions generated quickly.
- Data preprocessing worked correctly.
- Missing values handled properly.
- Hashes generated accurately.
- Ledger records stored securely.
- QR verification successful.
- Workflow transitions were smooth and consistent.

6.4 Validation

How Results Were Validated

The system outputs were validated using the following approaches:

1. **Prediction Verification**

- Compared prediction results generated by the model with expected outputs from the dataset.
- Verified that the system correctly classified heart disease risk based on the entered patient parameters.
- Checked that the prediction results were displayed accurately on the user interface.

2. Database Verification

- Checked PostgreSQL database records to confirm:
- Correct insertion of patient health parameters and prediction results.
- Proper storage of blockchain hash values in the ledger table.
- Accurate linking between patient records and prediction reports.

3. Blockchain Integrity Verification

- Verified that SHA-256 hashes were generated correctly for each prediction report.
- Confirmed that each ledger entry maintained proper hash chaining for tamper detection.
- Checked that report verification using QR code or transaction ID returned valid results.

4. End-to-End Workflow Testing

- Tested the complete system workflow:

  Patient Login → Enter Health Parameters → AI Prediction → Blockchain Hash Storage → Report Verification → Doctor Consultation
- Verified that prediction results were stored securely and displayed correctly.
- Confirmed that doctors could access patient reports and provide consultation when required.
- Ensured that the entire process executed smoothly without data loss or system errors.

CHAPTER 7

RESULTS AND PERFORMANCE ANALYSIS

7 Results and Performance Analysis

7.1 Output Analysis

The system generates predictions indicating whether a patient is likely to have heart disease, along with a corresponding risk category. Sample outputs can be represented in tabular or graphical form, showing how predicted risk correlates with input features like age, cholesterol, and blood pressure. Dashboards may include graphs illustrating the distribution of predictions and trends over time.

7.2 Performance Metrics

Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score on the test subset of the UCI Cleveland Heart Disease dataset. Additional metrics, like a confusion matrix, can be used to analyze false positives and false negatives. From a system perspective, average response time for prediction requests and ledger operations is also considered.

7.3 Comparative Analysis

A comparative analysis highlights differences between the existing system and the proposed system.

Feature	Existing System	Proposed System
Data Security	Centralized DB, vulnerable to tampering	Blockchain-style ledger with hash chaining, tamper-evident storage
Accessibility	Limited to physical visits or basic portals	Telemedicine interface with remote access to reports and consultations
Diagnosis	Manual assessment by doctors	AI-based Decision Tree prediction supporting early diagnosis
Data Integrity	Difficult to verify	Hash-based verification using QR code or transaction ID
Transparency	Low, limited audit trails	High, with traceable ledger entries and transaction history

Table no. 7.3.1 Comparing Features b/w Existing & Proposed Systems

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

8 Conclusion and Future Enhancements

8.1 Conclusion

What Was Achieved

The Smart Healthcare System successfully integrates Artificial Intelligence–based heart disease prediction with blockchain-secured medical record management and telemedicine features. The system allows patients to enter their health parameters through a web interface and receive an early prediction of heart disease risk using a trained machine learning model.

The generated prediction reports are stored securely using a blockchain-style ledger with SHA-256 hashing, ensuring that medical records remain tamper-proof and verifiable. The system also enables doctors to access patient reports and provide consultation, improving communication between patients and healthcare professionals.

This project demonstrates the practical implementation of a smart healthcare platform by combining machine learning, blockchain security, database management, and web technologies to improve early diagnosis and data reliability.

How Objectives Were Met

- The primary objectives of the project were successfully achieved:
- To develop a system that predicts heart disease risk using patient health parameters.
- To implement a machine learning model trained using the Kaggle Heart Disease dataset.
- To ensure secure storage of prediction reports using blockchain-based hashing techniques.
- To provide a verification mechanism using QR codes or transaction IDs.

  To enable doctors to review prediction reports and provide medical consultation through telemedicine features.

Key Outcomes

- Developed a functional AI-based heart disease prediction system.
- Integrated blockchain-style secure storage for medical reports.
- Implemented digital health record management for patients.
- Enabled report verification using QR code or transaction ID.
- Provided telemedicine support for doctor consultation.
- Demonstrated the feasibility of combining AI, blockchain, and web technologies in healthcare systems.

8.2 Limitations

Although the Smart Healthcare System performs effectively as a prototype, certain limitations exist.

Dataset Size

- The prediction model is trained using the Kaggle Heart Disease dataset, which contains a limited number of records.
- A larger and more diverse dataset could improve prediction accuracy and reliability.

System Environment

- The system is currently tested in a controlled development environment.
- Real-world deployment may require additional infrastructure and optimization.

Blockchain Implementation

- The blockchain component is implemented as a simplified ledger system for educational purposes.
- A fully distributed blockchain network is not implemented in the current version.

8.3 Future Scope / Enhancements

The Smart Healthcare System can be further improved with several enhancements

Additional Features

- Integration of real-time video consultation for telemedicine.
- Mobile application development for easier access to the system.
- Multi-language support for wider accessibility.
- Notification system for appointment reminders and health alerts.

Advanced Techniques

- Training the prediction model on larger and more diverse healthcare datasets.
- Implementation of advanced machine learning or deep learning models for improved prediction accuracy.
- Integration of wearable health device data for real-time monitoring.

Real-World Deployment Possibilities

- Deployment on cloud platforms such as AWS, Azure, or Google Cloud.
- Integration with hospital management systems and electronic health records (EHR).
- Development of a scalable architecture to support large numbers of users.

With further improvements and real-world deployment, the system has strong potential to support early disease detection, secure medical record management, and improved healthcare accessibility.

CHAPTER-9

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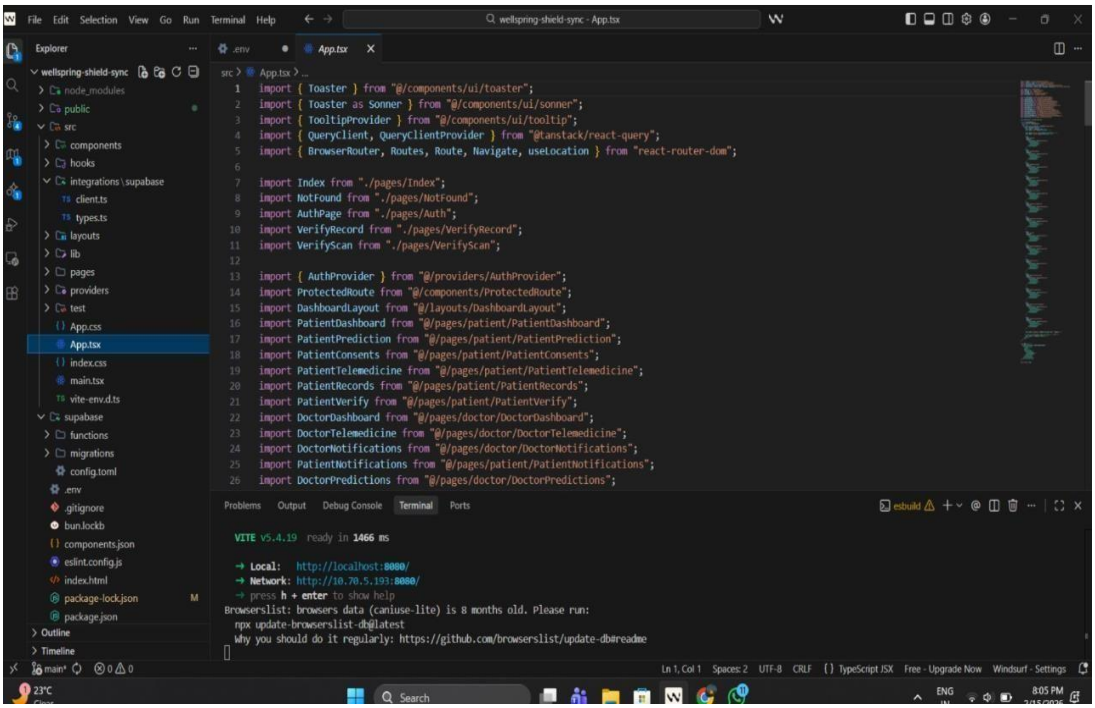
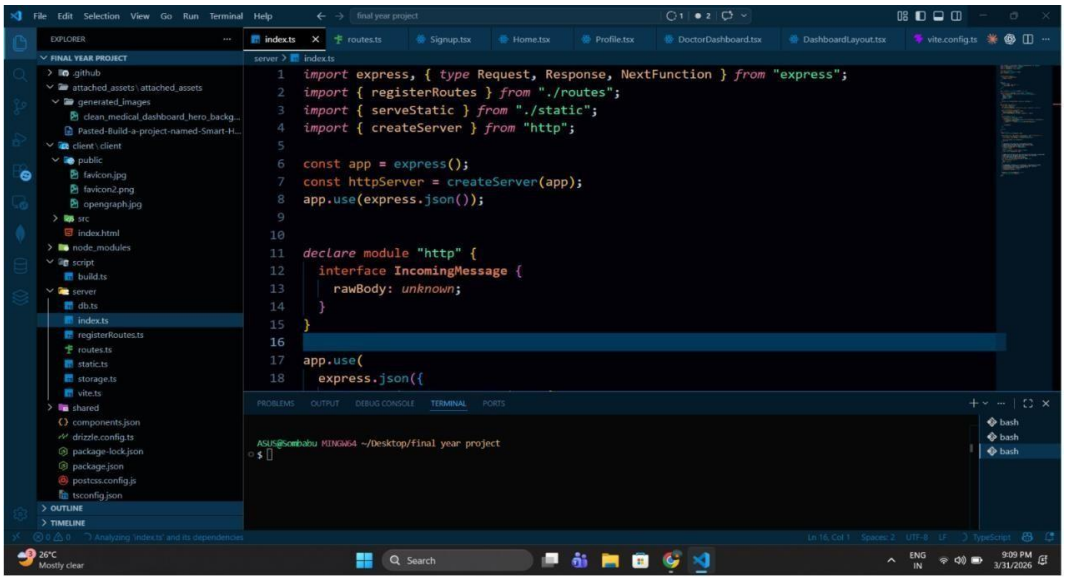
CHAPTER-10

APPENDIX

10 Appendix

10.1 Source Code (Sample / Snippets)





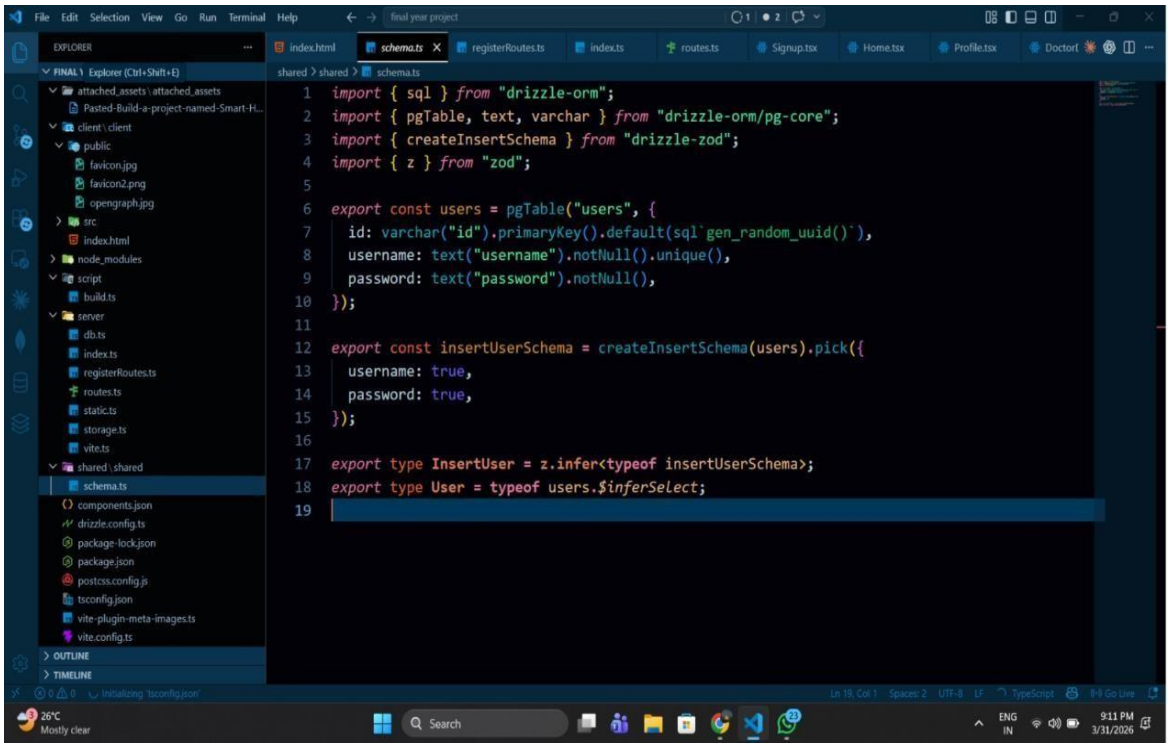


Fig no. 10.1.1 Source code

10.2 User Manual

Installation Steps

1. Install Python 3.x on your system.
2. Create a Supabase account and set up a new project.
3. Create the required database tables in Supabase.
4. Download or clone the project files.
5. Open the project folder in the terminal or command prompt.
6. Create a virtual environment. **- python -m venv venv**
7. Activate the virtual environment. **- venv\Scripts\activate**
8. Install the required dependencies. **- pip install -r requirements.txt**
9. Add Supabase project URL and API key in the project configuration file.

How to Run the Project

1. Ensure the Supabase database is active.
2. Open the terminal in the project directory.
3. Start the backend server. - **python manage.py runserver**
4. Open a web browser and access the application. - http://127.0.0.1:8000/

10.3 Abbreviations

- AI – Artificial Intelligence
- ML – Machine Learning
- API – Application Programming Interface
- DBMS – Database Management System
- UI – User Interface
- UCI – University of California, Irvine (dataset repository)
- QR – Quick Response (code)

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CardioShield AI: A Smart Blockchain-Enabled System for Early Heart Disease Detection and Secure Telemedicine

Sombabu Patel[1], P. Shakthi Vamsi[2], Sujal Sourav[3], T. Ayyappa Reddy[4], Mr Y. Durga Prasad[5]

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Abstract:

The artificial intelligence and blockchain-powered intelligent healthcare system is designed to improve the quality of early detection of heart diseases, improve the safety of medical records, and allow remote access to healthcare. The model is based on a decision tree classifier to examine the health parameters of the patients and predict cardiovascular risk. A blockchain-inspired ledger secures the prediction output through SHA-256 hashing and chaining, providing tamper-evident medical record storage. The presence of a combined telemedicine unit will allow patients and doctors to communicate with each other remotely via report access and consultations. The experimental results demonstrate the model's satisfaction and the system's suitability for use in a multi-user environment. The system proposed is more secure, transparent, and accessible than the traditional healthcare platforms.

Index Terms-

Smart Healthcare, Artificial Intelligence, Blockchain, Heart Disease Prediction, Decision Tree, Telemedicine, Secure Medical Records.

INTRODUCTION:

Most of the developing areas have healthcare systems that are characterized by delays in diagnosing conditions, poor records and access to specialist care. Most hospital management systems are based on centralized databases, which can be hacked, coerced or hit by a single point of failure. Besides, Heart disease is one of the leading causes of mortality in the world, and it is therefore important to have a quick diagnosis. There is a good potential in predictive analytics of healthcare using Artificial Intelligence (AI). Machine learning models are made to examine the characteristics of the patients, including age, cholesterol level, blood pressure and the type of chest pains, to predict the risk of disease. Nevertheless, prediction will not be sufficient when medical records lack security and verifiability. The cryptographic hashing and distributed records offer blockchain technology immutability and transparency. This project will result in an integrated smart health care ecosystem by combining AI-based prediction, blockchain-style secure storage, and telemedicine functionality.

LITERATURE SURVEY

Several studies use machine learning models, including logistic regression, Support Vector Machines, and Decision Trees, on data related to heart disease, like the UCI Cleveland data. The techniques are moderately predictive but, in most cases, may not be able to take into consideration medical record security. Healthcare blockchain applications are aimed at protected storage and decentralized access

control. Nonetheless, the majority of blockchain systems are achieved without AI-powered analytics. Telemedicine systems increase accessibility, although they mostly depend on central storage systems. The gap in the research is the integration of AI-based predictive analytics, a blockchain-ensured integrity of medical records, and an Internet-based telemedicine platform into a single architecture.

SYSTEM ARCHITECTURE

The system follows a three-layer architecture.

A. Presentation Layer

It was built in React.js and TypeScript and offers patient, doctor, and administrator role-based dashboards.

B. Application Layer

Functioning on Flask back-end API controls of AI prediction processing, authentication, ledger hash creation, and telemedicine processes.

C. Data Layer

Patient information and prediction history are stored in PostgreSQL. A table of ledger is of the blockchain fashion, where the hash of the data is made using SHA-256, which builds a secure hash chain to maintain the data integrity.

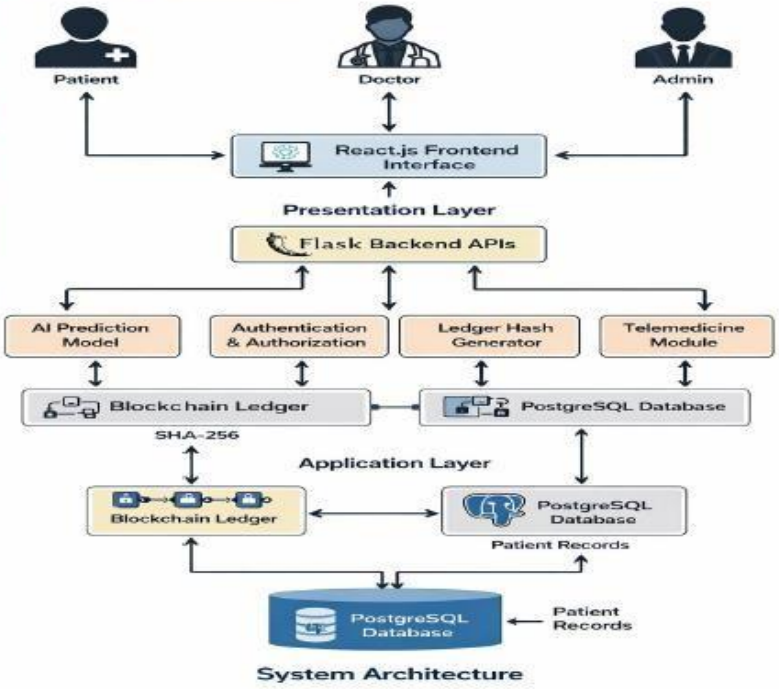


Fig. 1. Three-level system execution of clinical input to secure blockchain-based storage data flow

METHODOLOGY

A. Dataset and Preprocessing

The UCI Cleveland Heart Disease dataset is the dataset that will be used. Data preprocessing consists of managing the missing values, feature encoding, normalisation, and an 80:20 train-test split.

B. Decision Tree Model

Scikit-learn is used to train a Decision Tree classifier. As the splitting criterion, the Gini index is added. The model is trained to learn decision rules that will be used to classify the patients as either disease or no-disease cases.

C. Blockchain Ledger Mechanism

After prediction:

1) Report details are concatenated.
2) SHA-256 hash is generated.
3) The previous hash is appended.
4) Ledger entry is stored in the database.
5) This guarantees tamper-evident record storage.

D. Telemedicine Workflow

The patients can review reports, download QR codes, and consult. Doctors are able to retrieve reports on the patient, check their integrity by comparing hashes, and add clinical notes.

IMPLEMENTATION

A. Technologies Used

The implementation uses Python with Flask for REST APIs, React.js for the dynamic frontend, PostgreSQL for persistent storage, Scikit-learn for machine learning, and SHA-256 cryptographic hashing for security.

B. Algorithm Steps

The workflow begins with loading and preprocessing the dataset. The Decision Tree model is trained and deployed. Upon receiving patient input, the system predicts disease risk, generates a SHA-256 hash linked to the previous record, stores the ledger entry, and displays the result with QR verification.

RESULTS AND PERFORMANCE ANALYSIS

A. Performance Metrics

The Decision Tree model achieved acceptable accuracy, precision, recall, and F1-score on the test dataset.

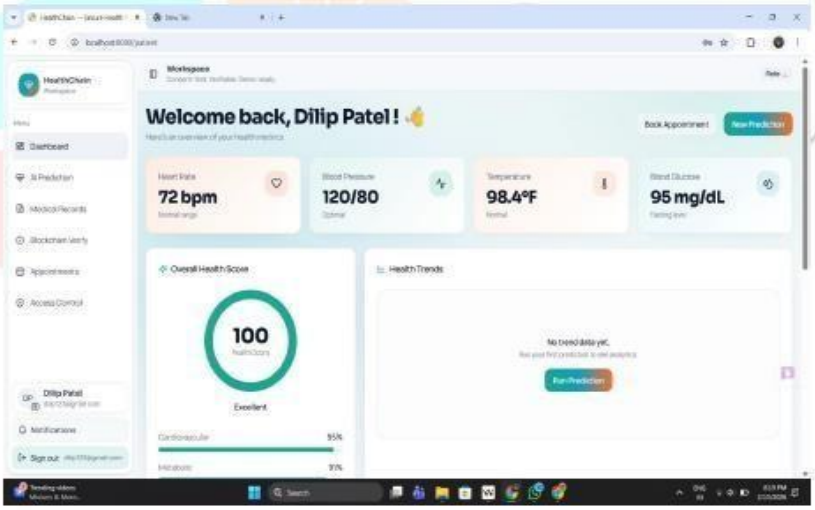
B. Confusion Matrix Analysis

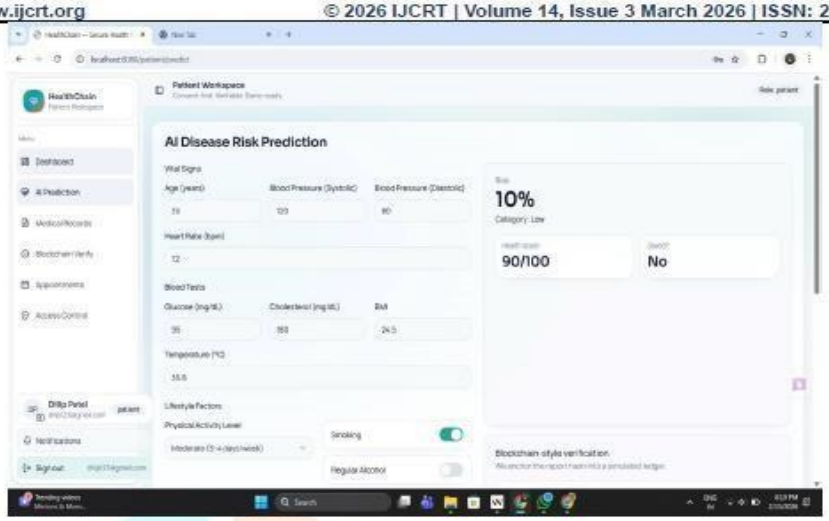
The confusion matrix indicates that the results are correctly classified in positive and negative cases. A prototype healthcare system can tolerate false positives and false negatives.

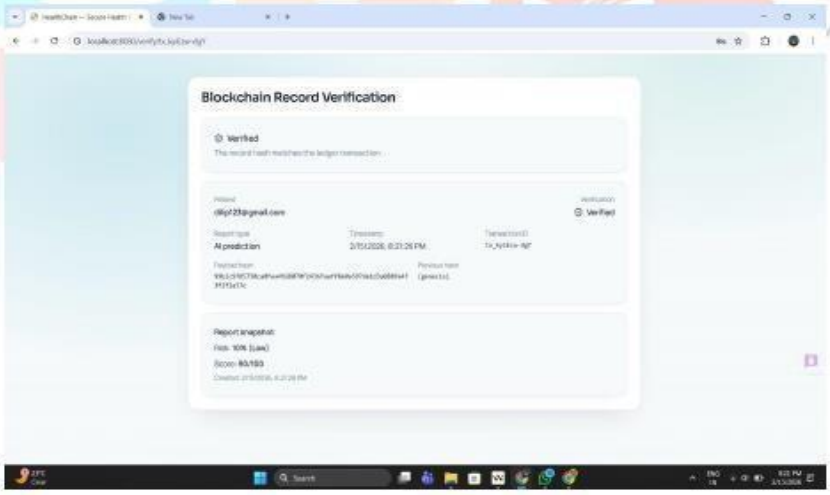
C. System Performance

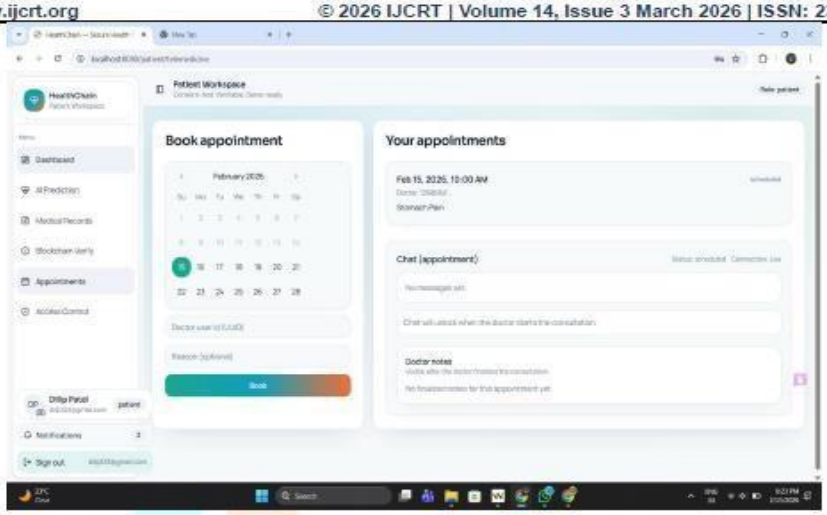
They had system benchmarks of an average prediction and ledger generation time of less than 2 seconds and less than 1 seconds respectively. The system was stable when tested under the conditions of multiple users.

HealthChain
Secure Health Records with Blockchain & AI
AI Disease Risk Prediction
Blockchain Record Verification
Consent-Based Access Control
Welcome Back
Sign In
Sign Up
Patient
Doctor
Email
Password
Sign In


HealthChain
Workspace
Dashboard
AI Prediction
Medical Records
Blockchain Verify
Appointments
Access Control
Dilip Patel
Notifications
Sign out
Welcome back, Dilip Patel!
Book Appointment
New Prediction
Heart Rate
72 bpm
Blood Pressure
120/80
Temperature
98.4°F
Blood Glucose
95 mg/dL
Overall Health Score
100
Excellent
Cardiovascular
95%
Metabolic
91%
Health Trends
No trend data yet.
Run Prediction


Patient Workspace
AI Disease Risk Prediction
10%
90/100
No


Blockchain Record Verification
Verified



COMPARATIVE ANALYSIS

This section compares the proposed Smart Healthcare System with traditional healthcare systems in terms of prediction capability, data security, verification mechanism, accessibility, and transparency.

COMPARISON OF EXISTING AND PROPOSED SYSTEMS

Feature	Existing	Proposed
Prediction	Basic ML	Decision Tree
Data Security	Central DB	Hash-chained Ledger
Verification	Limited	QR-based
Accessibility	Physical Visits	Telemedicine
Transparency	Low	Ledger Tracking

FUTURE SCOPE

Among the future enhancements, it is proposed to use larger datasets and deep learning models, full permissioned blockchain integration, live video consultation, and hospital management system integration.

CONCLUSION

Smart Healthcare System is an effective implementation of AI-based prediction of heart diseases and blockchain-style management of medical records. The classifier named Decision Tree is useful for early diagnosis; a hash-chained ledger helps to ensure tamper-evident storage. Telemedicine is more accessible and offers a safe, transparent alternative to traditional platforms.

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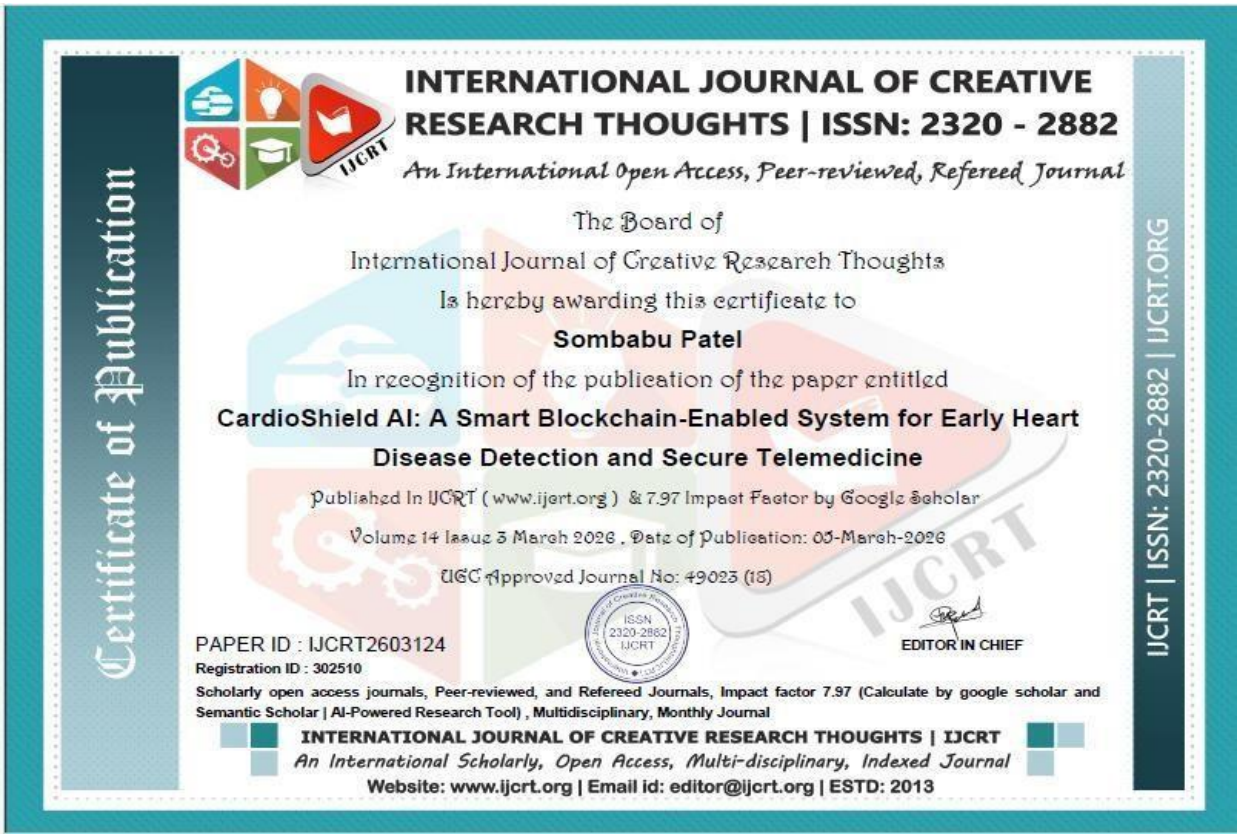

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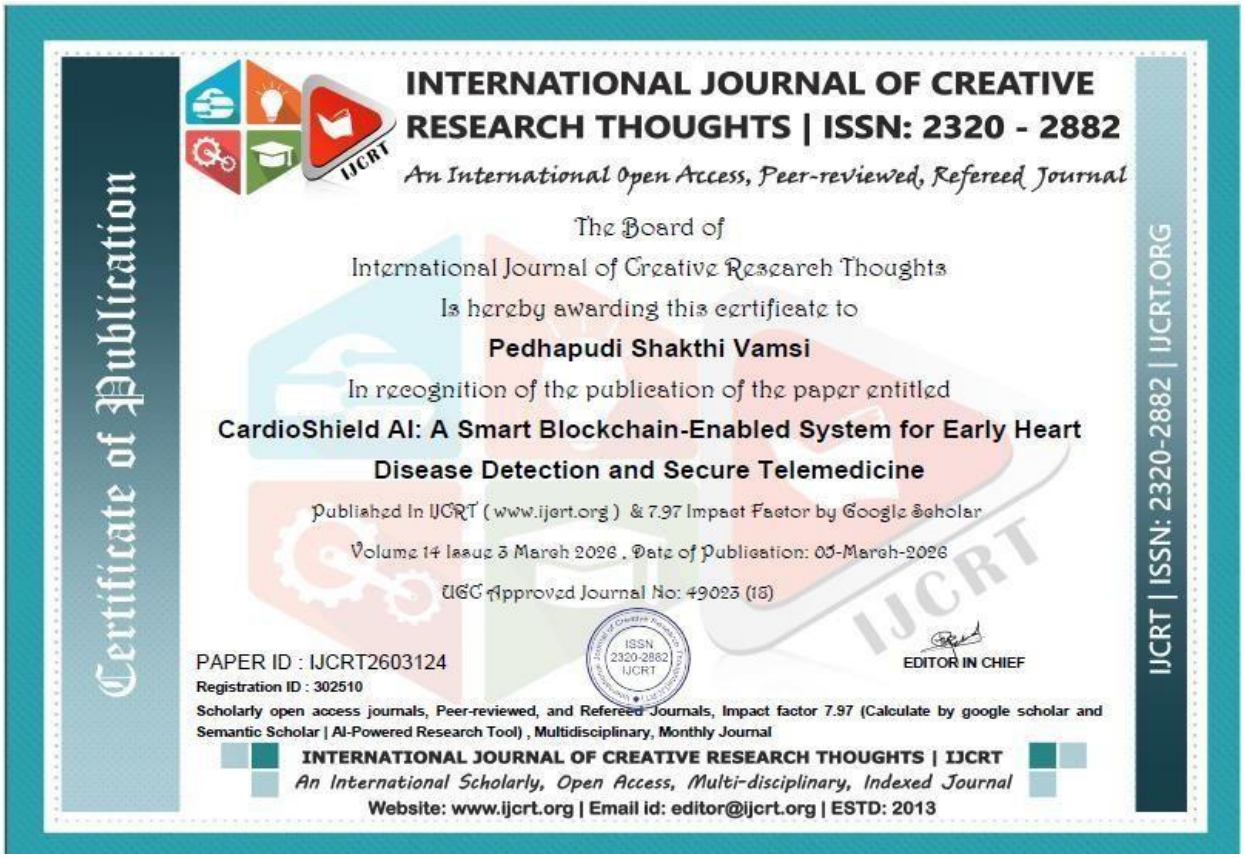

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